

Aggregation Semantics for Prioritization in Cooperative Automated Decision Making

Pavel Novoa-Hernández^{1,*} , Mariia Godz²  and David A. Pelta² 

¹ Dept. Computer Engineering and Systems, Universidad de La Laguna; pnovoah@ull.edu.es

² Dept. Computer Science and Artificial Intelligence, Universidad de Granada; {mariagodz,dpelta}@ugr.es

* Correspondence: pnovoah@ull.edu.es

Abstract

Cooperative automated decision-making systems increasingly combine predictive assessments with contextual knowledge to prioritize alternatives under resource constraints. Although the aggregation of these sources is often treated as a technical integration step, the choice of aggregation operator can determine whether contextual policies are preserved or overridden by predictive evidence. This paper investigates this issue in the context of CADEMAs-ML, focusing on the integration of a predictive risk score R and a contextual alignment score Q . We analyze three aggregation regimes—linear, weighted geometric, and minimum aggregation—from complementary theoretical, simulation, and application perspectives. The theoretical analysis characterizes the conditions under which contextual vetoes ($Q = 0$) are preserved through aggregation and shows how compensatory operators can violate such restrictions. Controlled Monte Carlo experiments then examine policy compliance, sensitivity to predictive-score uncertainty, ranking stability, and the effect of the intervention capacity on the resulting Top- K set. Finally, an employee-attrition case study based on a reproducible CADEMAs-ML workflow illustrates how changing only the aggregation semantics can alter the employees selected for intervention. The results show that aggregation is not a neutral implementation choice: under the conditions considered, zero-absorbing operators preserve contextual vetoes, whereas compensatory linear aggregation can admit vetoed alternatives, particularly when predictive evidence receives greater weight. Predictive uncertainty can further modify the prioritization and interact with the available intervention capacity without eliminating these structural differences. The study therefore highlights aggregation semantics as an explicit policy and governance choice in cooperative automated decision-making, rather than merely a numerical mechanism for combining scores.

Keywords: cooperative automated decision making; aggregation operators; multi-criteria decision making; context-aware decision support; prioritization

Received:

Revised:

Accepted:

Published:

Copyright: © 2026 by the authors.

Submitted to *Mathematics* for possible open access publication under the terms and conditions of the [Creative Commons Attribution \(CC BY\)](https://creativecommons.org/licenses/by/4.0/) license.

1. Introduction

Automated decision-making increasingly relies on the integration of heterogeneous sources of evidence. Predictive models can estimate the likelihood or severity of an outcome, while contextual knowledge can express organizational requirements, constraints, preferences, or policy considerations that are not necessarily captured by the predictive model itself. In cooperative automated decision-making settings, these sources may be produced by partially independent decision-making components and subsequently integrated into a common prioritization mechanism. The resulting priority scores can then be

used to allocate scarce resources, for example by selecting a fixed number of alternatives for intervention. In such settings, the integration mechanism becomes an essential part of the decision policy rather than a merely technical post-processing step.

This observation raises a fundamental question: *what happens when different aggregation semantics are used to integrate predictive and contextual evidence?* In particular, a contextual assessment may encode a hard or veto-like requirement, such that an alternative with $Q_i = 0$ should not be prioritized regardless of its predictive score R_i . A compensatory aggregation operator, however, may allow sufficiently large predictive evidence to offset a complete contextual deficiency. Conversely, non-compensatory operators may preserve the contextual restriction by assigning zero priority whenever the corresponding contextual condition is not satisfied. These alternatives can lead to substantially different rankings and, under a resource constraint, to different sets of alternatives receiving an intervention. Consequently, aggregation semantics can determine whether a policy encoded in the contextual layer is preserved or weakened during the final integration stage.

The issue becomes more relevant when predictive assessments are uncertain. Prediction errors can modify the relative ordering of alternatives and can therefore propagate beyond the score level into the actual intervention set. Moreover, the effect of such perturbations may depend on the aggregation operator, its parameters, and the available intervention capacity. A useful analysis must therefore consider not only the mathematical properties of the aggregation functions, but also their consequences for policy compliance, ranking stability, and resource-constrained Top-K selection.

Existing research provides a broad range of aggregation mechanisms in decision analysis and fuzzy multi-criteria decision-making, while cooperative automated decision-making frameworks provide architectures for combining multiple sources of decision evidence. However, the consequences of choosing a particular aggregation semantics for the preservation of contextual vetoes and the resulting prioritization behavior have not been systematically characterized in the CADEMAs-ML setting. In particular, there is a need to connect the structural properties of aggregation operators with observable decision-level consequences, including policy violations, sensitivity to predictive uncertainty, and changes in the alternatives selected under a fixed intervention budget.

This paper addresses this gap by analyzing the integration of predictive and contextual scores in CADEMAs-ML through three representative aggregation regimes: a compensatory linear operator, a weighted geometric operator, and a minimum operator. The analysis combines formal results with controlled Monte Carlo experiments and an employee-attribution case study. The theoretical analysis establishes conditions under which contextual vetoes are preserved or can be violated. The simulation study then examines policy compliance, the effect of predictive-score perturbations, ranking agreement, and the dependence of observed violations on the Top-K intervention capacity. Finally, the case study applies the same aggregation alternatives to a reproducible CADEMAs-ML workflow in which predictive and contextual scores are generated independently before being integrated into the final prioritization.

The main contributions of this work are as follows:

- We formally characterize the relationship between aggregation semantics and contextual veto preservation, identifying zero absorption as the key structural property for maintaining a zero contextual score through the integration stage.
- We provide a theoretical and empirical analysis of how compensatory and zero-absorbing aggregation operators affect policy compliance and resource-constrained prioritization under controlled population conditions.

- We investigate the sensitivity of the resulting prioritization to predictive uncertainty, including its effects on policy violations, ranking stability, inter-operator agreement, and the composition of Top-K intervention sets.
- We demonstrate the practical relevance of these differences through a reproducible employee-attrition case study, showing that changing only the aggregation semantics can alter the composition of a fixed intervention budget.

The remainder of the paper is organized as follows. Section 2 presents the background and related work, covering the foundations of cooperative automated decision-making, fuzzy multi-criteria decision-making, and aggregation theory, and formulates the research questions addressed in this study. Section 3 introduces the CADEMAs architecture and its CADEMAs-ML specialization. Section 4 presents the aggregation operators considered and their theoretical analysis in terms of structural properties and policy implications. Section 5 describes the controlled simulation experiments, their sensitivity analysis, and the employee-attrition case study. Section 6 discusses the implications of the findings, their relationship to aggregation theory and fuzzy MCDM, and the limitations of the study. Finally, Section 7 summarizes the conclusions and outlines directions for future research.

2. Background and Related Work

This section establishes the conceptual and methodological foundations of the study. It first introduces the cooperative automated decision-making setting in which predictive and contextual assessments are integrated, and then reviews the relevant foundations from fuzzy multi-criteria decision making and aggregation theory. Particular attention is given to aggregation properties that determine whether evidence from one decision pathway can compensate for deficiencies in another. The section concludes by positioning the present study with respect to existing CADEMAs-ML and aggregation-based approaches and by formulating the research questions that guide the subsequent theoretical, simulation, and case-study analyses.

2.1. Context-Aware and Cooperative Decision Support Systems

The increasing adoption of Artificial Intelligence (AI) in decision-support systems has highlighted the need to complement predictive performance with contextual, organizational, and domain-specific knowledge. In many real-world settings, decisions cannot be based solely on statistical predictions because institutional policies, expert knowledge, regulatory requirements, and operational constraints must also be considered. As a result, a growing body of research has focused on context-aware and cooperative decision-support systems capable of integrating heterogeneous sources of information into the decision-making process.

One important research direction concerns the incorporation of explicit knowledge representations alongside data-driven models. Early examples include cognitive-agent architectures that combine symbolic rules with machine-learning mechanisms to support context-aware decision making [1]. Similarly, ontology-based clinical decision-support systems have employed contextual ontologies and clinical pathways to select context-appropriate recommendations for individual cases [2]. More recent developments have extended these ideas through semantic models, knowledge graphs, and hybrid AI architectures that combine predictive models with structured domain knowledge to improve decision quality and interpretability [3,4].

A second research stream emphasizes cooperation between human and artificial intelligence. Rather than viewing decision support as a purely predictive task, these approaches consider decision making as a collaborative process in which machine-learning models provide evidence while human experts contribute contextual understanding, oversight,

and governance. Cronholm and Göbel [5], for example, argue that effective decision support requires appropriate mechanisms for combining human and artificial intelligence rather than simply improving predictive accuracy. Likewise, recent surveys on human-centred AI highlight transparency, accountability, contextual awareness, and governance as fundamental requirements for trustworthy decision-support systems [6].

Related ideas also appear in cooperative and distributed decision-making environments. Multi-agent systems, collective-intelligence frameworks, and cooperative AI architectures frequently rely on multiple decision sources that must be reconciled before a final recommendation is produced [7]. In these settings, contextual information often plays a coordinating role by constraining, guiding, or validating the outputs generated by predictive components. Similar concerns arise in governance-oriented decision-support systems, where contextual information may encode operational procedures, organizational priorities, safety requirements, or regulatory obligations [4,8].

Taken together, these studies clearly demonstrate the importance of contextual knowledge in decision-support systems. However, their primary focus typically lies in knowledge representation, ontology design, human–AI interaction, governance mechanisms, or system architecture. Once predictive evidence and contextual assessments have been generated, the numerical integration process that combines both sources of information is often treated as a secondary implementation detail. Consequently, relatively little attention has been devoted to understanding how alternative integration mechanisms affect the final recommendations produced by the system.

This limitation becomes particularly visible in architectures that explicitly separate predictive evidence from contextual assessment. Recent work on Cooperative Automated Decision-Making Systems (CADEMAS) [9] and its machine-learning instantiation CADEMAs-ML [10] exemplifies this design philosophy. CADEMAs organizes heterogeneous automated decision-making components and contextual knowledge into distinct layers, allowing predictive evidence and contextual assessments to be generated independently before being combined into a final recommendation. Such architectures make the integration mechanism itself an explicit design choice rather than an implicit implementation detail. As a result, the aggregation operator determines how predictive evidence interacts with contextual requirements, organizational priorities, or policy constraints, thereby shaping the semantics of the resulting decision process.

This observation suggests that aggregation should not be viewed merely as a numerical post-processing step. Instead, it can be interpreted as a governance mechanism that regulates the interaction between predictive and contextual information. Understanding the implications of different aggregation semantics therefore becomes essential for the design of cooperative and context-aware decision-support systems. The next subsection reviews the aggregation literature from this perspective.

2.2. Aggregation Operators and Decision Semantics

Aggregation operators constitute one of the central building blocks of multi-criteria decision making (MCDM), fuzzy decision support, and collective decision-making systems. Their purpose is to combine multiple sources of information into a single assessment while preserving, to a greater or lesser extent, the decision semantics associated with the original criteria. Consequently, aggregation has been studied extensively from both theoretical and applied perspectives.

Among the most influential aggregation frameworks are Ordered Weighted Averaging (OWA) operators [11,12], t-norm and t-conorm based operators, geometric aggregation functions, and non-additive integrals such as the Choquet and Sugeno integrals. Collectively, these approaches provide a rich mathematical vocabulary for representing different

attitudes toward compensation, conjunction, disjunction, and interaction among criteria. From this perspective, the choice of aggregation operator is not merely a numerical convenience but a modeling decision that determines how favorable and unfavorable evidence may compensate for one another.

A substantial body of literature has explored these properties in traditional decision-making settings. Reviews by Mardani et al. [13] and Csiszár [14] highlight the widespread use of OWA-based aggregation and related fuzzy operators across diverse application domains. More recent studies have continued to investigate how aggregation semantics influence consensus formation, robustness, criterion interaction, and ranking behavior. Examples include consensus-oriented aggregation in group decision making [15,16], robust Choquet-based decision models [17,18], and applications employing geometric and OWA-type operators to capture alternative decision attitudes [19,20].

Despite this methodological richness, most aggregation studies are formulated within classical MCDM settings, where the aggregated inputs represent multiple criteria, attributes, preferences, or expert judgments associated with the same decision problem. The primary objective is typically to identify the most appropriate operator for combining these criteria or to develop new aggregation mechanisms with desirable mathematical properties. Comparatively less attention has been paid to architectures in which the aggregated inputs originate from fundamentally different decision layers.

This distinction is particularly important in cooperative automated decision-making systems. In architectures such as CADEMAs-ML, the aggregation process does not combine multiple criteria describing an alternative. Instead, it reconciles two conceptually distinct sources of evidence: a predictive assessment generated by machine-learning models and a contextual assessment encoding organizational priorities, regulatory requirements, expert knowledge, or policy constraints. Under this interpretation, the aggregation operator effectively determines the extent to which predictive evidence may compensate for weak contextual support, or conversely, whether contextual constraints can restrict otherwise favorable predictions. The operator therefore acts as an implicit governance mechanism rather than merely a criterion-combination tool.

For this reason, the present work focuses on three representative aggregation regimes exhibiting fundamentally different compensation semantics. The weighted arithmetic mean represents a fully compensatory integration strategy in which predictive and contextual information may offset one another. The weighted geometric mean represents a partially compensatory strategy that penalizes strong imbalances between both sources of evidence. Finally, the minimum operator represents a non-compensatory strategy in which the weakest component determines the final assessment. Together, these operators span a spectrum ranging from highly permissive to highly restrictive integration behavior while remaining interpretable and widely understood within the aggregation literature.

From the perspective of this study, the objective is therefore not to propose a new aggregation operator nor to extend aggregation theory. Instead, aggregation operators are treated as alternative decision semantics within a cooperative automated decision-making architecture. The key question is how different compensation regimes influence policy compliance, prioritization behavior, robustness to predictive perturbations, and ranking stability when predictive and contextual information must jointly determine the final recommendation.

2.3. Research Gap and Positioning

The reviewed literature reveals two largely independent research streams. On the one hand, context-aware and cooperative decision-support systems have developed increasingly sophisticated mechanisms for incorporating contextual knowledge into predictive

processes through ontologies, symbolic rules, knowledge graphs, human oversight, and governance-oriented architectures [1,2,5,6]. These studies demonstrate the importance of contextual information but typically focus on knowledge representation, rule selection, system architecture, or human–AI interaction. Once predictive and contextual assessments have been generated, the mechanism responsible for their numerical integration is often treated as an implementation detail.

On the other hand, aggregation theory and fuzzy MCDM have developed a rich collection of operators with diverse compensation semantics, including OWA operators, geometric means, t-norms, Choquet integrals, and Sugeno integrals [11–14]. This literature provides extensive theoretical knowledge regarding compensation, criterion interaction, robustness, consensus formation, and preference aggregation. However, aggregation operators are generally studied as mathematical constructs or as components of multicriteria decision methods, where the aggregated inputs represent criteria, preferences, or expert judgments associated with the same decision problem.

Consequently, a gap exists between context-aware decision-support frameworks and aggregation theory. The former provides mechanisms for representing contextual knowledge but rarely investigates how different aggregation semantics affect the preservation of contextual constraints after integration. The latter offers powerful aggregation mechanisms but seldom studies their implications when heterogeneous decision signals, such as predictive machine-learning outputs and contextual assessments, must jointly determine operational recommendations.

From this perspective, the gap addressed in this work is not the absence of aggregation operators, nor the lack of mathematical knowledge about their properties. Rather, it is the lack of a systematic understanding of aggregation as a governance mechanism within cooperative automated decision-making architectures. In particular, relatively little attention has been devoted to questions such as whether contextual assessments can effectively act as veto mechanisms, how different compensation regimes influence policy compliance, how sensitive prioritization outcomes are to predictive overconfidence, or how ranking stability is affected by uncertainty in contextual information.

These observations motivate the following research questions:

- *RQ1. How do different aggregation semantics affect the interaction between predictive evidence and contextual assessments in a cooperative automated decision-making architecture?*
- *RQ2. Under what conditions can aggregation operators preserve or violate contextual constraints and veto-like policies?*
- *RQ3. How do alternative aggregation regimes influence prioritization robustness under predictive overconfidence and contextual uncertainty?*
- *RQ4. To what extent do different aggregation semantics alter the stability of the resulting rankings and intervention priorities?*

Unlike traditional fuzzy MCDM studies, the objective of the present work is not to propose a new aggregation operator, derive new algebraic results, or solve a classical multicriteria decision problem. Likewise, unlike many context-aware DSS frameworks, the focus is not on developing new mechanisms for representing contextual knowledge. Instead, aggregation operators are interpreted as alternative policy semantics within the CADEMAs architecture. By keeping both the predictive subsystem and the contextual subsystem fixed, the study isolates the aggregation mechanism as an independent design variable and examines its implications for policy compliance, robustness, and prioritization behavior. This positioning distinguishes the contribution from both aggregation theory and context-aware decision-support research.

Table 1. Chronological comparison of representative related work. “Partial” indicates that a contribution addresses the dimension only indirectly (e.g., through rules, consensus thresholds, or architectural design) without a dedicated analysis of predictive–contextual score fusion.

Reference	Year	Main focus	Predictive + contextual integration	Policy / veto analysis	Aggregation robustness	Cooperative ADM
Teng and Tan [1]	2008	Context-aware cognitive agents	✓	Partial	No	Partial
Yao and Kumar [2]	2012	Ontology-based clinical DSS	✓	Partial	No	No
Merkert et al. [21]	2015	ML in DSS survey	Partial	No	No	No
Moral et al. [15]	2017	Consensus aggregation	No	No	Partial	No
Mardani et al. [13]	2018	Fuzzy MCDM review	No	No	No	No
Csiszár [14]	2021	OWA theory	No	Partial	Partial	No
Cronholm and Göbel [5]	2022	Human-centred AI DSS	✓	Partial	No	Partial
Yang et al. [17]	2022	Robust Choquet-based MCDM	No	Partial	✓	No
Liao et al. [18]	2023	Contextual Choquet models	Partial	No	✓	No
Trillo et al. [16]	2023	Consensus-based GDM	No	Partial	No	No
Fanfani et al. [6]	2025	Human-centred AI survey	✓	Partial	No	Partial
Present work	2026	CADEMAS aggregation semantics	✓	✓	✓	✓

2.4. Chronological Comparison of Related Work

Table 1 summarizes representative contributions discussed in the previous subsections and positions the present work with respect to the existing literature. The comparison focuses on four dimensions that emerge from the identified research gap: (i) the explicit integration of predictive and contextual information, (ii) the treatment of contextual constraints or veto-like policies, (iii) the analysis of aggregation robustness, and (iv) the study of aggregation within cooperative automated decision-making architectures.

The table illustrates that prior research has typically focused either on context-aware decision-support architectures or on aggregation methods in MCDM and fuzzy decision making. Comparatively little attention has been devoted to aggregation as a governance mechanism that simultaneously integrates predictive evidence, contextual knowledge, policy constraints, and robustness considerations within a cooperative ADM framework.

3. CADEMAs and CADEMAs-ML

The Cooperative Automated Decision-Making System (CADEMAS) framework was introduced by Novoa-Hernández et al. [9] as a general architecture for context-aware decision support in which multiple heterogeneous Automated Decision-Making (ADM) units cooperate to produce a unified recommendation, score, or decision. Rather than prescribing

a specific decision model, CADEMAs provides a modular structure that separates the generation of decision evidence from its contextual evaluation and final integration.

This separation is particularly relevant for the problem studied in the present work. Many decision-support systems combine predictive evidence with contextual information, but these elements are often intertwined within the same model or decision process. In contrast, CADEMAs explicitly distinguishes three conceptually different functions: (i) evidence generation, (ii) contextual assessment, and (iii) decision integration. Consequently, the aggregation mechanism can be analysed independently of both the predictive subsystem and the contextual knowledge representation.

Formally, a CADEMAs instance is represented by the tuple

$$S = \langle I, U, O, K, T \rangle, \quad (1)$$

where I denotes the Input Manager, U the set of ADM units, O the Decision Integration Layer, K the Contextual Modulator, and T the universe of admissible data representations.

Given an input case, the ADM units independently generate decision evidence from the available information. In parallel, the Contextual Modulator evaluates the extent to which the same case satisfies organizational priorities, expert knowledge, regulatory requirements, or other contextual constraints. The outputs of both subsystems are then combined by the Decision Integration Layer to produce the final decision score.

An important specialization of this framework is CADEMAs-ML [10], in which the ADM units correspond to machine-learning models. In this setting, multiple predictive models cooperate to produce a consolidated predictive assessment, while contextual information is represented through a fuzzy inference system inspired by Pérez-Cañedo et al. [22] and Pérez-Cañedo et al. [23]. The contextual subsystem translates organizational priorities and institutional requirements into a normalized degree of contextual satisfaction.

To focus on the role of aggregation, the internal implementation of both subsystems is abstracted in terms of two normalized scores. For each decision case i , the predictive subsystem produces a predictive score

$$R_i \in [0, 1], \quad (2)$$

which summarizes the cooperative evidence generated by the predictive ADM units. Depending on the application domain, R_i may represent a risk estimate, a confidence score, a probability of occurrence, or any other normalized predictive assessment.

Independently, the contextual subsystem produces a context score

$$Q_i \in [0, 1], \quad (3)$$

which quantifies the degree to which the case satisfies the contextual requirements encoded by the organization, institution, or application domain.

It is worth noting that the restriction of both predictive and contextual scores to the unit interval does not limit the generality of the proposed formulation. Any continuous score defined over an interval $[a, b]$ can be transformed into the normalized domain $[0, 1]$ through an affine mapping. This transformation preserves ordering relations and therefore maintains the ranking semantics required in prioritization tasks. Consequently, R_i and Q_i should be interpreted as normalized semantic scores rather than measurements expressed in a physical unit. The use of the unit interval allows the analysis of aggregation properties independently of the original scale of the predictive and contextual information. More importantly, the lower boundary $Q_i = 0$ represents a complete *contextual violation*.

The final prioritization score is then obtained through an aggregation operator

$$P_i = A(R_i, Q_i), \quad (4)$$

where A defines the integration semantics between predictive evidence and contextual assessment. Candidate cases are subsequently ranked according to P_i , and the corresponding Top-K subset is selected for intervention or resource allocation.

It is important to emphasize that the aggregation problem considered here differs from traditional fuzzy MCDM settings. In classical MCDM, aggregation operators combine multiple criteria describing different attributes of the same alternative, typically with the objective of capturing trade-offs among decision criteria. In CADEMAs-ML, however, the operator combines heterogeneous sources of decision evidence that play fundamentally different roles: a predictive assessment generated by cooperating machine-learning models and a contextual assessment derived from organizational, regulatory, or domain-specific knowledge.

Consequently, the aggregation operator does not merely determine the relative importance of criteria. Instead, it governs how predictive evidence interacts with contextual requirements and therefore acts as an explicit policy-design component within the decision architecture. The central question is not which operator produces the most desirable aggregate score in a multicriteria sense, but rather how alternative aggregation semantics affect policy compliance, contextual veto preservation, robustness to predictive overconfidence, and ranking stability under uncertainty.

Existing CADEMAs-ML applications employ a weighted linear combination of predictive and contextual scores [10]. However, the CADEMAs framework itself does not prescribe any particular aggregation operator [9]. Consequently, different aggregation semantics can be introduced without modifying either the predictive subsystem or the contextual subsystem. This modularity provides an ideal setting for studying aggregation as a policy-design mechanism and for analysing how different compensation regimes affect prioritization, policy compliance, robustness, and ranking stability. The following section introduces the aggregation operators considered in this study and develops the theoretical properties that motivate the subsequent experimental analysis.

4. Aggregation Semantics and Theoretical Analysis

The aggregation mechanism in CADEMAs-ML determines how predictive evidence and contextual knowledge are integrated into a unified prioritization decision. Therefore, the aggregation operator should not be interpreted as a mere mathematical component, but as an explicit design choice that defines the interaction between machine-learning evidence and contextual requirements.

This section provides the theoretical foundations required to answer the research questions introduced in Section 2.3, particularly those related to the role of aggregation semantics in cooperative automated decision-making. Specifically, the analysis addresses how different aggregation regimes modify the interaction between predictive and contextual information (RQ1), under which conditions contextual constraints and veto-like policies are preserved or violated (RQ2), how aggregation semantics influence robustness to predictive overconfidence and contextual uncertainty (RQ3), and how they affect the stability of the resulting prioritization rankings (RQ4).

The analysis does not aim to introduce a new aggregation operator or to extend classical aggregation theory. Instead, it studies existing aggregation paradigms as alternative policy semantics within the CADEMAs-ML architecture. The section is organized in three steps. It first formalizes the structural properties that characterize aggregation behavior, including boundedness, monotonicity, continuity, zero absorption, contextual veto preservation, and compensability. It then examines three representative aggregation regimes with

different decision semantics: fully compensatory linear aggregation, sub-compensatory aggregation through the weighted geometric mean, and non-compensatory minimum aggregation. It closes by deriving conditions for contextual veto preservation, rank reversal under compensation, and sensitivity to perturbations in predictive and contextual assessments.

These results provide a formal basis for interpreting aggregation operators as policy-design components in cooperative automated decision-making systems and motivate the experimental evaluation of their practical consequences in the following sections.

4.1. Structural Properties of Aggregation Operators

Without loss of generality, all scores are assumed to be normalized within the unit interval. Any bounded continuous scoring scale $[a, b]$ can be transformed into this representation through a monotonic affine transformation, preserving the ordering relations and the aggregation semantics considered in this work.

Let

$$A : [0, 1]^2 \rightarrow [0, 1] \quad (5)$$

denote an aggregation operator that combines the predictive score R_i and the contextual score Q_i into the final prioritization score

$$P_i = A(R_i, Q_i). \quad (6)$$

Throughout this work, aggregation operators are interpreted as integration policies governing the interaction between predictive evidence and contextual knowledge within the CADEMAs-ML architecture. Rather than focusing on specific functional forms, the following definitions introduce a set of structural properties that characterize different aggregation semantics.

Definition 1 (Boundedness). *An aggregation operator is bounded if*

$$0 \leq A(r, q) \leq 1, \quad \forall (r, q) \in [0, 1]^2. \quad (7)$$

Boundedness guarantees that the resulting prioritization score remains within the same normalized scale as the predictive and contextual assessments. Since A maps $[0, 1]^2$ into $[0, 1]$, this property is automatically satisfied and is stated here only for completeness.

Definition 2 (Monotonicity). *An aggregation operator is monotonic if*

$$r_1 \leq r_2, q_1 \leq q_2 \implies A(r_1, q_1) \leq A(r_2, q_2). \quad (8)$$

Monotonicity ensures that improving either predictive evidence or contextual satisfaction cannot decrease the final prioritization score.

Definition 3 (Continuity). *An aggregation operator is continuous if small perturbations in its inputs produce arbitrarily small changes in the aggregated score.*

Continuity is particularly relevant when predictive and contextual scores are estimated from noisy data, since it prevents abrupt changes in prioritization due to infinitesimal variations of the inputs.

Definition 4 (Zero Absorption). *An aggregation operator satisfies the zero-absorption property if*

$$A(r, 0) = 0, \quad \forall r \in [0, 1]. \quad (9)$$

Under zero absorption, a null contextual assessment completely suppresses the final prioritization score regardless of the predictive evidence available.

Definition 5 (Strict positivity on the interior). *An aggregation operator is strictly positive on the interior if*

$$A(r, q) > 0, \quad \forall (r, q) \in (0, 1]^2. \quad (10)$$

Zero absorption and strict positivity play complementary roles. The first forces a null score exactly when the context vetoes an alternative. The second guarantees that no other alternative is driven to zero. Taken together, they make the value $P_i = 0$ an unambiguous marker of a contextual veto, which is what allows veto preservation to be read directly from the aggregated scores.

Definition 6 (Contextual Veto Preservation). *An aggregation operator preserves contextual vetoes if*

$$Q_i = 0 \implies P_i = 0. \quad (11)$$

Within the CADEMAs-ML setting, contextual veto preservation means that contextual requirements cannot be overridden by predictive evidence. Observe that zero absorption is a sufficient condition for contextual veto preservation. **In fact, for $Q_i = 0$, zero absorption and contextual veto preservation are equivalent.**

Definition 7 (Compensability). *In this work, an aggregation operator is compensatory on an interior region if a decrease in one input can be locally offset by an increase in the other while preserving the aggregated score. For a differentiable operator, a sufficient local characterization is that both partial derivatives are strictly positive in that region.*

Thus, compensability denotes local substitutability rather than the ability to offset an arbitrary deficit. The unit-scale bounds limit every trade-off. It determines the extent to which strong predictive evidence may compensate for weak contextual support, or vice versa. At one extreme, fully compensatory operators permit the broadest within-scale trade-offs between prediction and context. At the opposite extreme, non-compensatory operators prevent such trade-offs and therefore behave as hard contextual constraints.

These properties provide a theoretical basis for distinguishing alternative integration semantics within CADEMAs-ML. In particular, zero absorption and compensability play a central role in determining whether contextual requirements behave as soft preferences, partial constraints, or strict vetoes. The following subsection introduces the specific aggregation operators considered in this work and analyzes how they instantiate these structural properties.

4.2. Aggregation Operators Considered

The CADEMAs framework does not prescribe a specific mechanism for combining predictive and contextual assessments [9]. Consequently, different aggregation operators can be incorporated without modifying either the predictive subsystem or the contextual subsystem.

To investigate how aggregation semantics affect prioritization behavior, policy compliance, and robustness, this work considers three representative operators that span different levels of compensability between predictive evidence and contextual requirements.

1. *Linear aggregation.* The linear operator corresponds to the integration mechanism adopted in existing CADEMAs-ML implementations [10]:

$$A_L(R_i, Q_i) = \lambda R_i + (1 - \lambda)Q_i, \quad \lambda \in [0, 1]. \quad (12)$$

The parameter λ controls the relative importance assigned to predictive and contextual information. For $\lambda \in (0, 1)$, this operator is compensatory: reductions in one component can be offset by increases in the other, subject to the bounds of the unit interval. The boundary cases $\lambda = 0$ and $\lambda = 1$ are degenerate and are not considered as compensatory regimes in the subsequent analysis.

2. *Weighted geometric aggregation.* An intermediate alternative is the weighted geometric mean:

$$A_G(R_i, Q_i) = R_i^\lambda Q_i^{1-\lambda}, \quad \lambda \in (0, 1). \quad (13)$$

Unlike the linear operator, the weighted geometric mean penalizes imbalanced evaluations and therefore reduces the degree of compensation between predictive and contextual information. Nevertheless, partial compensation remains possible whenever both inputs are strictly positive. Restricting the exponent to $\lambda \in (0, 1)$ keeps both powers strictly between 0 and 1, so the indeterminate form 0^0 never arises and $A_G(r, 0) = 0$ holds for every r . The two excluded endpoints behave differently: at $\lambda = 1$ the operator reduces to $A_G(R_i, Q_i) = R_i$ and loses the veto property, and at $\lambda = 0$ it reduces to Q_i . Both are therefore excluded from the analysis, and the implementation assigns $A_G = 0$ whenever either input equals zero.

3. *Minimum aggregation.* The third regime is the minimum operator

$$A_M(R_i, Q_i) = \min(R_i, Q_i). \quad (14)$$

This operator represents a non-compensatory integration regime in which the final prioritization score cannot exceed the weakest component. Consequently, predictive evidence cannot compensate for insufficient contextual support. Note that A_M carries no weighting parameter. Its output is invariant under changes of λ , so any comparison across integration regimes leaves the minimum operator unchanged by construction. This point matters when reading the λ -dependent results of Sections 5.2 and 5.3.

Table 2 summarizes the structural properties of the three operators according to the definitions introduced in the previous subsection.

Table 2. Structural properties of the aggregation operators considered in this work.

Property	Linear	Geometric	Minimum
Boundedness	Yes	Yes	Yes
Monotonicity	Yes	Yes	Yes
Continuity	Yes	Yes	Yes
Zero absorption	No [†]	Yes	Yes
Contextual veto preservation	No [†]	Yes	Yes
Strict positivity on $(0, 1]^2$	Yes	Yes	Yes
Compensability	Full [‡]	Partial	No

[†] For the linear operator, zero absorption and contextual veto preservation hold only in the degenerate case $\lambda = 0$.

[‡] Full compensability for the linear operator is understood for $\lambda \in (0, 1)$, with compensation bounded by the $[0, 1]$ scale.

All three operators are strictly positive on the interior, so they differ only in zero absorption. This isolates the property responsible for the contrasting policy behaviour reported in Section 5: the operators agree on which alternatives deserve a positive score and disagree only on whether a contextual veto can be overridden.

These operators were selected because they represent three distinct integration semantics within CADEMAs-ML: full within-scale compensation, partial compensation, and strict contextual constraint. The following subsections analyze the theoretical consequences of these semantics and their implications for prioritization behavior. Here, full within-scale compensation refers to the linear operator with $\lambda \in (0, 1)$; it does not mean unbounded compensation.

4.3. Conditions for Contextual Veto Preservation

Within CADEMAs-ML, contextual information may represent different types of requirements, including organizational priorities, regulatory constraints, safety conditions, or expert-defined restrictions. In some applications, a contextual assessment should not behave as a soft preference but rather as a hard constraint. This situation occurs when a complete contextual violation must prevent an alternative from being prioritized, regardless of its predictive evidence.

This behavior can be characterized through the zero-absorption property introduced in Section 4.1. An aggregation operator provides contextual veto preservation if

$$A(r, 0) = 0, \quad \forall r \in [0, 1]. \quad (15)$$

Therefore, the question of whether CADEMAs-ML can enforce contextual vetoes reduces to determining whether the selected aggregation operator satisfies zero absorption.

Theorem 1 (Contextual veto preservation). *Let $A : [0, 1]^2 \rightarrow [0, 1]$ be an aggregation operator used in CADEMAs-ML. If A satisfies zero absorption, then every decision case i with a null contextual assessment,*

$$Q_i = 0, \quad (16)$$

receives a null prioritization score,

$$P_i = A(R_i, Q_i) = 0, \quad (17)$$

independently of its predictive score R_i .

Proof. The result follows directly from the definition of zero absorption. Since $Q_i = 0$, it follows that $P_i = A(R_i, Q_i) = A(R_i, 0)$. Because A satisfies zero absorption, $A(R_i, 0) = 0$, for every $R_i \in [0, 1]$. Therefore, $P_i = 0$. Hence, a complete contextual violation completely suppresses the final prioritization score regardless of the predictive evidence. $\square$

Corollary 1 (Veto preservation of the considered operators). *Among the aggregation operators considered in this work, the minimum and weighted geometric operators preserve contextual vetoes, whereas the linear operator does not for $\lambda \in (0, 1]$, with the degenerate case $\lambda = 0$ being an exception.*

Proof. For the minimum operator, $A_M(r, 0) = \min(r, 0) = 0$. Similarly, for the weighted geometric operator, $A_G(r, 0) = r^\lambda 0^{1-\lambda} = 0$, for $\lambda \in (0, 1)$. In contrast, for $\lambda \in (0, 1]$, the linear operator yields $A_L(r, 0) = \lambda r$, which is greater than zero whenever $r > 0$ and $\lambda > 0$. Therefore, predictive evidence can override a complete contextual violation under linear aggregation. For the degenerate case $\lambda = 0$, $A_L(r, 0) = 0$, so the linear operator trivially preserves contextual vetoes. $\square$

Theorem 1 constrains individual prioritization scores. In a Top- K selection, however, the relevant question is whether these score-level guarantees are sufficient to exclude vetoed alternatives from the selected set. The following corollary makes this connection explicit and identifies the additional population-level condition required.

Let

$$\mathcal{V} = \{i : Q_i = 0\}$$

denote the set of vetoed alternatives, and let $\mathcal{T}_K$ denote the set of alternatives selected among the K largest prioritization scores.

Corollary 2 (Exclusion of vetoed alternatives from the Top- K set). *Let A satisfy zero absorption and strict positivity on the interior (Definition 5). Consider a population of N alternatives, of which N_V are vetoed ($Q_i = 0$), while every remaining alternative satisfies $R_i > 0$ and $Q_i > 0$. If*

$$N - N_V \geq K,$$

then

$$\mathcal{T}_K \cap \mathcal{V} = \emptyset.$$

Proof. By zero absorption, every vetoed alternative receives a zero prioritization score,

$$P_i = A(R_i, 0) = 0, \quad i \in \mathcal{V}.$$

By strict positivity on the interior, every non-vetoed alternative satisfying $R_i > 0$ and $Q_i > 0$ receives a strictly positive score,

$$P_i = A(R_i, Q_i) > 0, \quad i \notin \mathcal{V}.$$

Since $N - N_V \geq K$, there are at least K non-vetoed alternatives with strictly positive prioritization scores. Consequently, the K largest scores are all strictly positive, whereas every vetoed alternative has score zero. Hence no vetoed alternative can belong to the Top- K set, and therefore

$$\mathcal{T}_K \cap \mathcal{V} = \emptyset.$$

$\square$

The corollary shows that zero absorption provides a score-level guarantee that extends to Top- K selection when a sufficient number of admissible alternatives is available. Thus, zero absorption alone is not sufficient to characterize the selected set: a population-level supply condition, $N - N_V \geq K$, is also required. If fewer than K non-vetoed alternatives are available, some vetoed alternatives may necessarily remain in the selected set, even though the aggregation operator assigns them a zero prioritization score.

The previous result establishes that contextual veto preservation is not a consequence of the predictive model or the contextual knowledge representation, but rather an emergent property of the integration semantics. Consequently, two CADEMAs-ML systems with identical predictive and contextual components may exhibit fundamentally different policy behavior depending only on the selected aggregation operator.

This observation motivates the subsequent analysis of rank reversal conditions and compensation thresholds, which characterize how much contextual improvement is required to offset predictive advantages under different aggregation regimes.

4.4. Conditions for Rank Reversal under Compensation

The previous section established that contextual veto preservation depends on the existence of a zero-absorption property. However, many aggregation operators used in decision-support systems follow a different philosophy: they allow partial compensation between the aggregated sources of evidence. In such cases, a sufficiently high value in one dimension may offset a lower value in another dimension.

This property is common in compensatory aggregation schemes, including weighted arithmetic means and several non-linear aggregation operators. Although compensation provides flexibility when predictive evidence and contextual information represent partially substitutable criteria, it also introduces the possibility that the final prioritization ranking differs from the ranking induced by predictive evidence alone.

To formalize this phenomenon, let $A : [0, 1]^2 \rightarrow [0, 1]$ be a continuous aggregation operator that is strictly increasing in both arguments. Given two alternatives i and j , assume that alternative i has stronger predictive evidence, but alternative j has better contextual alignment: $R_i > R_j$ and $Q_i < Q_j$. The final prioritization scores are then $P_i = A(R_i, Q_i)$ and $P_j = A(R_j, Q_j)$. A rank reversal occurs when $P_j > P_i$, meaning that the contextual advantage of alternative j compensates for its predictive disadvantage.

Theorem 2 (General rank reversal condition under compensation). *Let $A : [0, 1]^2 \rightarrow [0, 1]$ be a continuous aggregation operator strictly increasing in both arguments. For two alternatives i and j satisfying predictive dominance $R_i > R_j$, assume the operator provides sufficient compensatory capacity such that there exists a contextual assessment $q_{\max} \leq 1$ satisfying $A(R_j, q_{\max}) > A(R_i, Q_i)$.*

Under these conditions, a rank reversal in favour of alternative j occurs if and only if $Q_j > Q_i + \Phi_A(R_i, R_j, Q_i)$, where Φ_A is the minimum contextual improvement required to compensate for the predictive difference, implicitly defined by $A(R_j, Q_i + \Phi_A) = A(R_i, Q_i)$.

Proof. Fix $R_i > R_j$ and Q_i . Define the difference function $g(q) = A(R_j, q) - A(R_i, Q_i)$. Because A is continuous and strictly increasing in its second argument, $g(q)$ is also continuous and strictly increasing. At $q = Q_i$, $g(Q_i) = A(R_j, Q_i) - A(R_i, Q_i) < 0$ because A is strictly increasing in its first argument and $R_j < R_i$. By hypothesis, there exists $q_{\max}$ such that $g(q_{\max}) > 0$. Therefore, by the Intermediate Value Theorem, there exists a unique contextual value $q^* \in (Q_i, q_{\max})$ such that $g(q^*) = 0$, which implies $A(R_j, q^*) = A(R_i, Q_i)$. Defining the required threshold as $\Phi_A(R_i, R_j, Q_i) = q^* - Q_i$, the relation $P_j > P_i$ (or $g(Q_j) > 0$) holds if and only if $Q_j > q^*$, which is equivalent to $Q_j > Q_i + \Phi_A(R_i, R_j, Q_i)$. $\square$

Under its stated feasibility condition, the theorem shows that rank reversal need not be an artefact of a particular aggregation formula: it can arise when predictive and contextual information are allowed to compensate for one another. The specific threshold Φ_A depends on the operator and on the two alternatives being compared. The theorem does not imply that every compensatory operator, or every pair of alternatives, admits a reversal within the unit interval.

This distinction is particularly relevant in CADEMAs-ML because the two aggregated components do not represent equivalent evaluation criteria. The predictive score R_i summarizes evidence generated by machine learning models, whereas the contextual score Q_i represents compliance with organizational, regulatory, or domain-specific requirements. Therefore, the choice of aggregation operator implicitly defines the degree to which contextual requirements can modify or override data-driven recommendations.

4.5. Sensitivity to Predictive and Contextual Perturbations

The previous sections analyzed the structural properties of aggregation operators in terms of compensation and contextual veto preservation. However, in practical decision-support systems, both predictive scores and contextual assessments are subject to uncertainty. Predictive models may produce unstable estimates due to limited data, model uncertainty, or distribution changes, whereas contextual assessments may vary because of incomplete knowledge, expert disagreement, or evolving organizational requirements.

Therefore, an additional relevant property of an aggregation mechanism is its sensitivity to perturbations in its input dimensions. In the context of CADEMAs-ML, this property determines how strongly changes in predictive evidence or contextual requirements affect the final prioritization score.

Let $A : [0, 1]^2 \rightarrow [0, 1]$ be an aggregation operator. Consider a perturbation ϵ applied to the contextual assessment while the predictive score remains fixed. The resulting variation in the prioritization score is $\Delta_Q(\epsilon) = |A(R, Q + \epsilon) - A(R, Q)|$. Similarly, a perturbation in the predictive assessment produces $\Delta_R(\epsilon) = |A(R + \epsilon, Q) - A(R, Q)|$. These quantities measure the local influence of each information source on the final decision score. Operators that strongly amplify small perturbations may lead to unstable rankings, whereas operators with bounded sensitivity provide greater robustness against uncertainty in predictive or contextual information.

Theorem 3 (Lipschitz sensitivity of aggregation operators). *Let $A : [0, 1]^2 \rightarrow [0, 1]$ be an aggregation operator continuously differentiable on an open neighbourhood containing the line segment between (R, Q) and $(R + \epsilon_R, Q + \epsilon_Q)$. Assume that both endpoints, and hence the whole segment by convexity, lie in $[0, 1]^2$. If there exist finite constants L_R and L_Q such that*

$$|\partial A / \partial R| \leq L_R \quad \text{and} \quad |\partial A / \partial Q| \leq L_Q,$$

then, for any perturbations ϵ_R and ϵ_Q ,

$$|A(R + \epsilon_R, Q + \epsilon_Q) - A(R, Q)| \leq L_R |\epsilon_R| + L_Q |\epsilon_Q|.$$

Proof. By the Mean Value Theorem, there exists a point (ξ_R, ξ_Q) between (R, Q) and $(R + \epsilon_R, Q + \epsilon_Q)$ such that

$$A(R + \epsilon_R, Q + \epsilon_Q) - A(R, Q) = (\partial A / \partial R)(\xi_R, \xi_Q) \epsilon_R + (\partial A / \partial Q)(\xi_R, \xi_Q) \epsilon_Q.$$

Taking absolute values and applying the derivative bounds yields

$$|A(R + \epsilon_R, Q + \epsilon_Q) - A(R, Q)| \leq L_R |\epsilon_R| + L_Q |\epsilon_Q|.$$

□

The theorem establishes a local robustness bound. The constants L_R and L_Q quantify the maximum amplification of perturbations on the particular region considered; they do not by themselves establish a global ordering of operators or the stability of a population-level Top-K decision. Those questions are examined empirically in Section 5.

For the linear aggregation operator, $A_L(R, Q) = \lambda R + (1 - \lambda)Q$, the sensitivity coefficients are constant: $L_R = \lambda$ and $L_Q = 1 - \lambda$. Therefore, the influence of predictive and contextual uncertainty is uniform throughout the decision space. This property explains the fully compensatory behaviour of the linear operator: a perturbation in one dimension has exactly the same marginal effect regardless of the current values of prediction and context.

For the weighted geometric operator, $A_G(R, Q) = R^\lambda Q^{1-\lambda}$, the local sensitivities are $\partial A_G / \partial R = \lambda R^{\lambda-1} Q^{1-\lambda}$ and $\partial A_G / \partial Q = (1 - \lambda) R^\lambda Q^{-\lambda}$. Unlike the linear operator, these sensitivities depend on the current decision state. In particular, $\partial A_G / \partial Q$ becomes unbounded as $Q \rightarrow 0^+$ when $R > 0$. Thus, finite Lipschitz constants for A_G are available only on regions bounded away from the coordinate axes. Near a contextual veto, a small positive observational error can therefore have a large local effect on the aggregated score. This fact is consistent with, but does not by itself determine, a false-negative-veto mechanism under imperfect contextual observation.

For the minimum operator, $A_M(R, Q) = \min(R, Q)$, the final score is completely determined by the limiting component. Consequently, the operator exhibits zero compensation: improving the non-limiting input has no effect until it becomes the active constraint. This property provides maximal preservation of contextual restrictions but may reduce responsiveness when contextual assessments are uncertain. Because A_M is not differentiable on $R = Q$, it is not covered by Theorem 3 there; the following corollary captures one of its piecewise-stable regimes.

Corollary 3 (Minimum operator stability under inactive-component perturbations). *Consider a decision case evaluated under the minimum operator $A_M(R, Q) = \min(R, Q)$. If the predictive score acts as the strict limiting component ($R < Q$), the operator completely absorbs any contextual perturbation ε as long as the ordinal relationship is maintained (i.e., $R \leq Q + \varepsilon$). Under these conditions, the sensitivity is zero:*

$$|A_M(R, Q + \varepsilon) - A_M(R, Q)| = |R - R| = 0.$$

Proof. Let the initial state of the decision case be defined by R and Q . By hypothesis, the predictive score is the strict limiting component, meaning $R < Q$. By the definition of the minimum operator, the initial aggregated score is:

$$A_M(R, Q) = \min(R, Q) = R.$$

Now, consider a perturbed contextual assessment $Q' = Q + \varepsilon$. By hypothesis, the perturbation ε (whether positive or negative) is bounded so that the ordinal relationship is maintained, that is, $R \leq Q + \varepsilon$. Evaluating the minimum operator under this perturbed state yields:

$$A_M(R, Q + \varepsilon) = \min(R, Q + \varepsilon) = R.$$

Calculating the absolute difference between the perturbed and initial states gives the sensitivity:

$$|A_M(R, Q + \varepsilon) - A_M(R, Q)| = |R - R| = 0.$$

Thus, the perturbation is completely absorbed, and the final prioritization score remains unaffected. $\square$

This result illustrates the asymmetric robustness of non-compensatory operators. Perturbations of the non-limiting component are absorbed entirely, whereas perturbations of the active constraint pass through to the aggregated score. The transmission is never amplified: since $|\min(R, Q + \epsilon) - \min(R, Q)| \leq |\epsilon|$ for all R, Q, ϵ , the minimum operator is globally 1-Lipschitz in each argument, and its sensitivity therefore alternates between 0 and at most $|\epsilon|$. The contrast with A_G is worth noting: the geometric operator is smooth but its contextual sensitivity grows without bound as $Q \rightarrow 0^+$, whereas A_M is non-smooth yet uniformly bounded. Smoothness and robustness are thus not equivalent here. These are score-level bounds, however, and they do not by themselves determine how stable the induced ranking is; that question is addressed empirically in Section 5.3.

These analytical differences motivate the robustness experiments conducted in the following sections. Rather than evaluating aggregation operators only according to final ranking performance, the experiments examine how their semantic properties influence stability under perturbations of predictive evidence and contextual assessments.

5. Simulation Results

The theoretical analysis establishes how the aggregation operators differ in terms of compensation, veto preservation, and sensitivity to perturbations. The simulation study complements these score-level results by examining their consequences when the aggregated scores are used to prioritize a population of competing alternatives. The experiments therefore focus on two aspects that cannot be determined from the operator definitions alone: how the resulting prioritization changes as the relative weight of predictive and contextual evidence varies, and how stable the induced rankings and Top-K selections remain under perturbations of the underlying scores.

The experimental design described below uses controlled Monte Carlo populations to isolate these effects while keeping the decision architecture fixed. The results are organized accordingly: first, we examine the λ -sensitivity of the three aggregation semantics, with particular attention to the compensatory behavior of the linear and geometric operators; second, we evaluate the stability of the induced rankings and Top-K selections under perturbations of the predictive score R . This latter analysis deliberately treats R as the uncertain input, reflecting the predictive nature of the ADM outputs in the CADEMAs-ML setting. Although contextual assessments Q could likewise be uncertain in other implementations, a separate Q -noise experiment is unnecessary here: the aggregation operators exhibit the corresponding argument-wise properties derived theoretically, so the conclusions obtained for perturbations of R extend directly to perturbations of Q under the analogous conditions. Throughout the analysis, the empirical findings are therefore interpreted in relation to the theoretical properties established in Section 4.

5.1. Experimental Design

The simulation study uses controlled Monte Carlo populations to isolate the effects of aggregation semantics while keeping the underlying decision architecture fixed. Each population contains $N = 1000$ alternatives, with each alternative i characterized by a predictive score $R_i \in [0, 1]$ and a contextual assessment $Q_i \in [0, 1]$, representing the outputs of the decision and contextual pathways, respectively. The same population is evaluated under the linear, weighted geometric, and minimum aggregation operators,

$$P_i = A(R_i, Q_i; \lambda),$$

which determines the resulting prioritization and Top-K selection. Unless otherwise stated, $K = 100$ and 20% of the population is assigned a contextual veto ($Q_i = 0$). Vetoed alternatives are generated with comparatively high predictive scores, creating explicit conflicts between predictive evidence and contextual admissibility. The remaining scores are sampled from bounded distributions designed to provide both aligned and conflicting combinations of R_i and Q_i , including weak-context cases with $0 < Q_i \leq 0.25$.

The first analysis performs a dense sweep of $\lambda \in [0, 1]$ to characterize how the relative contribution of predictive and contextual evidence modifies the resulting prioritization. The linear and weighted geometric operators are evaluated throughout this sweep, whereas the minimum operator is independent of λ and therefore provides a natural reference. For the complementary comparison of prioritization stability, we use seven representative aggregation configurations: $A_L^{0.10}$, $A_L^{0.50}$, $A_L^{0.90}$, $A_G^{0.10}$, $A_G^{0.50}$, $A_G^{0.90}$, and A_M . The three λ values represent low, balanced, and high relative weight on predictive evidence, respectively, while avoiding an unnecessarily large number of pairwise comparisons.

All experiments use 1000 Monte Carlo replications with registered seeds 42–1041. The evaluation considers three complementary aspects of the resulting prioritization: contextual policy compliance, predictive utility, and decision stability. Let $\mathcal{T}_K$ denote the selected Top-K set and let

$$\mathcal{V} = \{i : Q_i = 0\}$$

denote the set of vetoed alternatives. Policy compliance is measured by the policy violation rate

$$V = \frac{|\mathcal{T}_K \cap \mathcal{V}|}{K},$$

which represents the proportion of selected alternatives that violate a contextual veto. The metric counts only strict vetoes ($Q_i = 0$) and does not treat weak non-veto cases ($0 < Q_i \leq 0.25$) as violations. Thus, $V = 0$ corresponds to complete preservation of the contextual vetoes, whereas larger values indicate greater policy non-compliance. Because K is fixed and positive, V is well defined for every experimental condition.

Predictive utility is characterized by the mean original predictive score of the selected alternatives,

$$\bar{R} = \frac{1}{K} \sum_{i \in \mathcal{T}_K} R_i.$$

The average is computed over the complete Top-K set, including vetoed alternatives when present. This keeps the denominator fixed and allows $\bar{R}$ to be interpreted alongside V : the former describes the predictive evidence represented in the selected set, whereas the latter describes its contextual compliance. In the perturbation experiments, $\bar{R}$ is computed from the original, unperturbed R_i values so that the metric measures the predictive quality of the selected alternatives rather than the magnitude of the injected noise.

The second analysis evaluates robustness to uncertainty in the predictive pathway by perturbing the predictive score according to

$$R'_i = \text{clip}(R_i + \epsilon_i, 0, 1), \quad \epsilon_i \sim \mathcal{N}(0, \sigma_R^2), \quad (18)$$

and comparing the resulting prioritization with the corresponding noiseless reference. We focus the uncertainty analysis on R because it represents the output of an ADM or predictive model and is therefore naturally subject to estimation error. No separate Q -noise experiment is required: the corresponding score-level sensitivity properties extend to Q by the symmetry of the aggregation operators. The clipping operation is retained because both scores are bounded to $[0, 1]$.

Decision stability is assessed from two complementary perspectives. For a reference aggregated-score vector $P^{(1)}$ and a perturbed or alternative vector $P^{(2)}$, Kendall's τ_b measures agreement between their global rankings, with the tie correction inherent to the statistic. In contrast, the Jaccard similarity measures agreement between the corresponding Top- K sets,

$$J = \frac{|\mathcal{T}_K^{(1)} \cap \mathcal{T}_K^{(2)}|}{|\mathcal{T}_K^{(1)} \cup \mathcal{T}_K^{(2)}|}.$$

Kendall's τ_b therefore captures changes distributed across the complete ranking, whereas J focuses directly on the stability of the alternatives that would actually be prioritized. For Top- K selection, ties in P are resolved using the predictive score available in the corresponding condition as a secondary criterion and the alternative identifier as a final deterministic criterion, ensuring reproducible selections.

Monte Carlo means are reported together with 95% percentile intervals, which describe the dispersion of outcomes across independently generated populations rather than the sampling uncertainty of the estimated mean. The simulation study is therefore intended to characterize the population-level consequences of the analytically established aggregation properties, rather than to reproduce their proofs.

5.2. Policy Compliance under Contextual Vetoes

The first experiment examines the population-level consequences of the contextual veto property established in Section 4.3. It addresses RQ2 by asking when the aggregation semantics preserve a contextual exclusion and when predictive evidence can compensate for it. The focus is therefore not on overall predictive performance, but on how the aggregation rule translates the two information sources into a prioritization.

A controlled stress population was constructed with $N = 1000$ alternatives, of which 20% are contextual vetoes, represented by $Q_i = 0$. These alternatives are deliberately assigned comparatively high predictive scores, creating a direct conflict between predictive evidence and contextual admissibility. The remaining alternatives cover both aligned and conflicting combinations of R_i and Q_i . A dense sweep of $\lambda \in [0, 1]$ is used for the linear and weighted geometric operators, while the minimum operator is included as a reference because it is independent of λ . For every value of λ , 1000 independent Monte Carlo populations are generated and the resulting Top- K selections are evaluated with $K = 100$. Since 800 non-vetoed alternatives remain, the population satisfies the supply condition $N - N_V \geq K$ required by Corollary 2.

Figure 1(a) gives the main population-level result. For low values of λ , all three operators produce zero observed policy violations. As predictive evidence receives more weight, however, the linear operator departs from this behaviour. Its violation rate becomes positive at approximately $\lambda = 0.7$ and increases thereafter, reaching $V = 0.64$ at $\lambda = 1.0$ (95% percentile interval $[0.55, 0.73]$). The weighted geometric and minimum operators remain at $V = 0$ throughout the sweep. The onset around $\lambda = 0.7$ is specific to the simulated population and should therefore not be interpreted as a universal threshold for the linear operator.

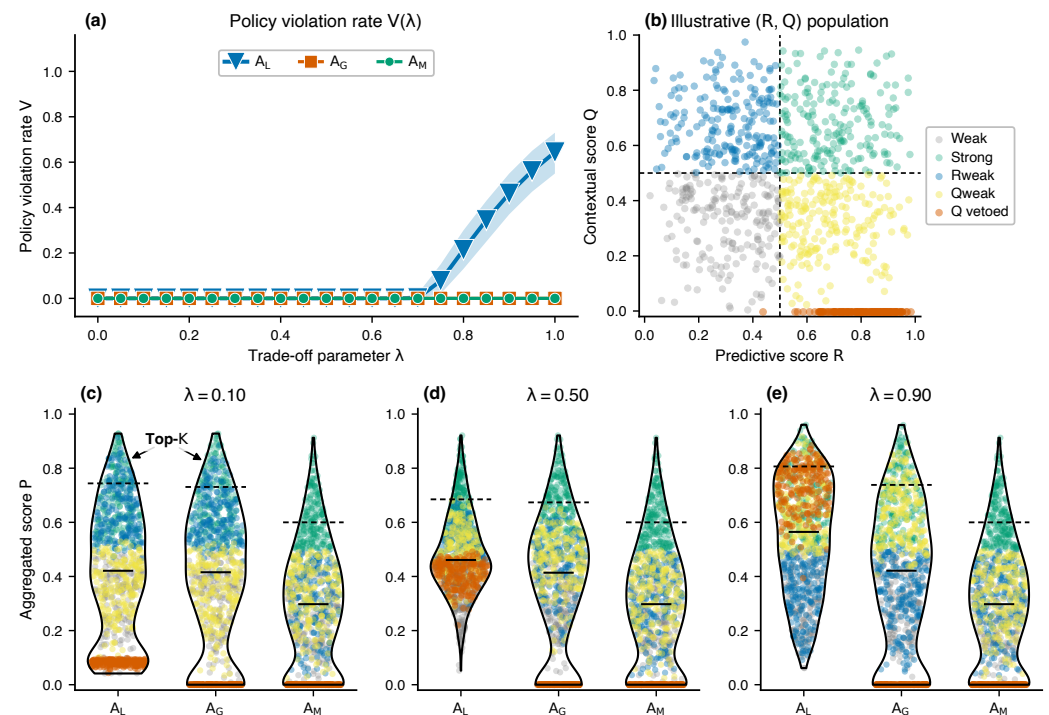


Figure 1. Policy compliance under contextual vetoes ($N = 1000$, $K = 100$). (a) Monte Carlo mean policy violation rate over $\lambda \in [0, 1]$ across 1000 replications; shaded bands indicate 95% percentile intervals. (b) Illustrative population in the (R, Q) plane, including contextually vetoed cases ($Q_i = 0$) and non-veto cases with different predictive–contextual profiles. (c–e) Aggregated-score distributions for A_L , A_G , and A_M at $\lambda = 0.10$, 0.50 , and 0.90 , respectively, for the same population. The dashed line indicates the Top- K cutoff and the overlaid points identify vetoed cases.

The remaining panels help trace this population-level pattern back to the individual scores. Panel (b) shows the first Monte Carlo population in the (R, Q) plane. The lines at $R = 0.5$ and $Q = 0.5$ separate broadly aligned cases, where predictive and contextual evidence point in the same direction, from conflicting cases, where one source is relatively strong while the other is weak. The vetoed alternatives form the horizontal $Q_i = 0$ band in the high- R region. This makes the intended stress condition visible directly: the cases that should be excluded are also among those that prediction would otherwise favour.

Panels (c)–(e) follow these same alternatives after aggregation. The dashed horizontal line marks the score of the K -th selected alternative, so points above it are competitive for Top- K selection. At $\lambda = 0.10$ and 0.50 , the vetoed cases remain outside this region for the linear operator, although their scores are already spread above zero. At $\lambda = 0.90$, their scores have moved sufficiently upward for some of them to reach the Top- K boundary. The weighted geometric and minimum operators show a different pattern: their vetoed cases remain at $P_i = 0$ independently of λ , while the non-vetoed population is redistributed above this boundary. The figure therefore makes visible the mechanism behind the transition observed in panel (a): violations arise when a compensatory aggregation rule allows predictive evidence to lift a vetoed alternative into competition with admissible alternatives.

This distinction follows directly from the theoretical results. By Theorem 1, zero absorption implies that $Q_i = 0$ is propagated to a zero aggregated score. Corollary 2 then adds the population-level requirement needed to turn this score-level property into Top- K exclusion. Both conditions hold in the present design for A_G and A_M , explaining why their zero violation rate is deterministic rather than an empirical indication of superior performance. The linear operator does not satisfy zero absorption, so the theory provides

no corresponding exclusion guarantee. Its behaviour instead depends on the distribution of predictive and contextual scores, which is precisely what the dense sweep quantifies.

The consequences extend beyond explicit vetoes to the final prioritization. Figure 2 compares the rankings and Top-K sets produced by seven representative configurations: A_L and A_G at $\lambda \in \{0.10, 0.50, 0.90\}$, together with A_M . Kendall's τ_b measures agreement between the complete aggregated-score rankings, whereas Jaccard similarity measures the overlap between the selected Top-K sets. At balanced weighting, the compensatory operators with same λ values remain relatively close: for example, $A_L(0.50)$ and $A_G(0.50)$ have mean $\tau_b > 0.88$ (Fig. 2-(a)). In contrast, configurations that combine different aggregation semantics and strongly different predictive weights can diverge substantially. The comparison between $A_L(0.90)$ and $A_G(0.10)$, for instance, gives mean $\tau_b < 0$ and a Jaccard overlap below 0.15. Thus, the choice of aggregation semantics can change not only whether a contextual veto is preserved, but also which alternatives are ultimately prioritized.

These agreement results should not be interpreted as a ranking of the operators themselves. They show instead that aggregation semantics define different mappings from the same (R_i, Q_i) evidence to the final decision order. The compliance results identify one particularly consequential manifestation of this difference: under a compensatory rule, increasing the predictive weight can make contextually excluded alternatives competitive; under zero-absorbing rules, the theoretical exclusion property prevents this transition under the stated population conditions. Together, the two analyses connect the operator-level properties established theoretically with their population-level consequences for both policy compliance and the resulting Top-K prioritization.

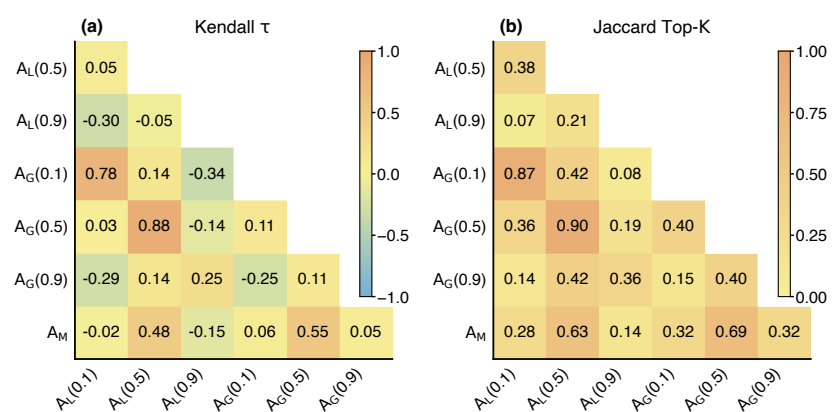


Figure 2. Pairwise agreement among the seven aggregation configurations considered in Experiment 01 ($N = 1000$, $K = 100$, 1000 Monte Carlo replications). The configurations are A_L and A_G at $\lambda \in \{0.10, 0.50, 0.90\}$ and A_M . Each cell reports the Monte Carlo mean of the corresponding pairwise statistic across replications. (a) Kendall's τ_b between the aggregated-score rankings. (b) Jaccard similarity between the corresponding Top-K sets. Higher values indicate greater agreement; the diagonal is omitted because self-comparisons are identically equal to one.

5.3. Sensitivity Analysis

The second experiment examines how uncertainty in the predictive pathway propagates to the prioritization produced by CADEMAs-ML, addressing RQ3 and RQ4. We focus on perturbations of the predictive score because R represents the output of an ADM or predictive model and can therefore be affected by estimation error. For the aggregation operators considered here, the corresponding analysis for contextual uncertainty is obtained by exchanging R and Q and, for the λ -dependent operators, replacing λ by $1 - \lambda$. We therefore use perturbations of R as the representative uncertainty channel rather than duplicating an analytically symmetric experiment on Q .

For each of the 1000 Monte Carlo populations, the predictive scores are perturbed according to Eq. (18), with $\sigma_R \in \{0.05, 0.10, 0.20\}$. The clean case $\sigma_R = 0$ is retained as a baseline. The contextual scores remain unchanged, including the 20% of cases with $Q_i = 0$. We evaluate the seven ranking configurations used throughout the experiments: A_L and A_G at $\lambda \in \{0.10, 0.50, 0.90\}$, together with A_M . These settings provide low, intermediate, and high predictive weighting for the two λ -dependent operators, while A_M serves as the λ -independent reference. For each configuration, we record the policy violation rate V and compare the noisy ranking with the clean ranking obtained from the same population before perturbation, using Kendall's τ_b and Top-K Jaccard similarity. We also compare the seven configurations under the same noisy realizations to distinguish changes relative to their own clean rankings from changes in agreement between aggregation semantics.

Figure 3 first illustrates the effect of predictive perturbations on the underlying population. Panels (a)–(c) show one representative Monte Carlo population at increasing noise levels. Because Q remains fixed, the perturbation acts horizontally in the (R, Q) plane. Cases subject to the contextual veto therefore remain on the $Q = 0$ boundary, whereas non-vetoed cases become increasingly dispersed along the predictive dimension. At the largest noise level, the clipping operation also produces visible accumulation at the boundaries of the admissible interval. Thus, the perturbation is not an unconstrained Gaussian displacement, but a noisy predictive score projected back onto $[0, 1]$.

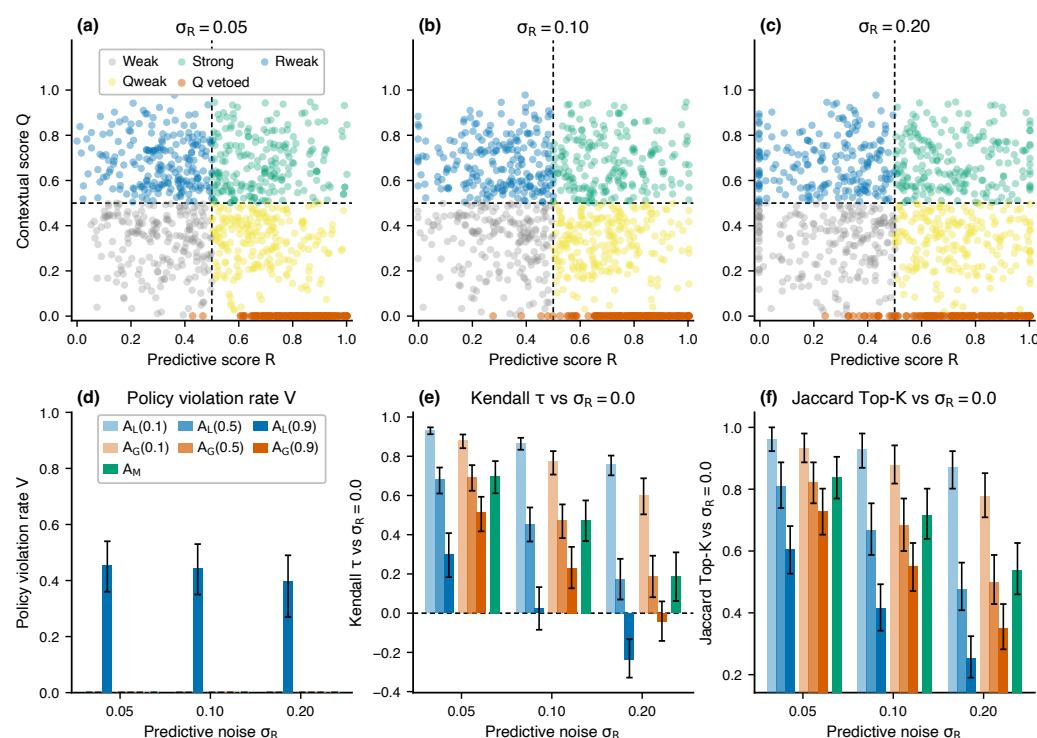


Figure 3. Sensitivity to predictive-score perturbations. Panels (a)–(c) illustrate one representative population at increasing noise levels, while panels (d)–(f) summarize the corresponding Monte Carlo effects on the seven ranking configurations.

The population-level ranking consequences are summarized by policy compliance and ranking stability. The zero-absorbing operators A_G and A_M retain $V = 0$ across the tested noise levels, whereas perturbing R does not provide additional protection for the compensatory linear operator. In particular, the high-predictive configuration $A_L(0.90)$ continues to admit policy-violating cases. This follows directly from Theorem 1: perturbing R does not alter the property that A_G and A_M map a true contextual veto $Q_i = 0$ to zero.

Since the population satisfies the supply condition of Corollary 2, this score-level property continues to exclude vetoed alternatives from the Top- K set.

Predictive uncertainty also reduces agreement with the corresponding clean rankings. At $\sigma_R = 0.05$, most configurations exhibit a moderate degradation, which becomes substantially stronger as the noise increases. The effect is configuration-dependent, with settings that assign greater weight to R generally being more sensitive, particularly for the linear operator. At $\sigma_R = 0.20$, for example, the mean Kendall correlation for $A_L(0.90)$ falls to approximately -0.24 , while its Top- K Jaccard similarity is only about 0.25. Thus, large predictive perturbations can substantially alter both the global ordering and the selected intervention set. The stronger degradation in Jaccard than in Kendall's τ_b further indicates that the instability is particularly consequential around the Top- K boundary.

The agreement between aggregation configurations under shared noisy realizations is shown in Figure 4. Unlike the previous analysis, which compares each noisy ranking with its own clean counterpart, these heatmaps compare different aggregation semantics under the same perturbed predictive scores. Agreement generally decreases as σ_R increases, although the magnitude of the effect varies across configurations. In particular, settings with different λ values can diverge substantially, whereas configurations with similar integration semantics tend to remain more closely aligned. Predictive uncertainty therefore does not erase the effect of aggregation semantics; rather, it interacts with the way each operator transforms the perturbed evidence into prioritization scores.

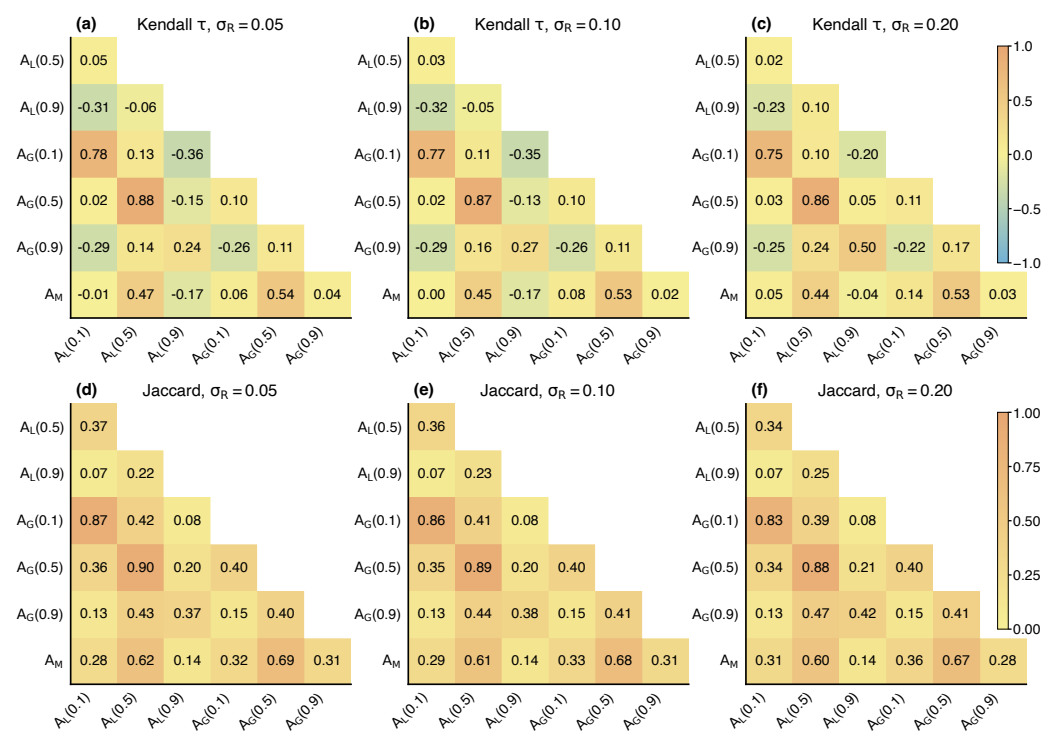


Figure 4. Pairwise agreement among the seven ranking configurations under shared predictive-score perturbations. The heatmaps report mean Kendall τ_b and Top- K Jaccard similarity across Monte Carlo replications.

A complementary analysis examines how the intervention capacity K interacts with predictive uncertainty. Figure 5 reports the mean policy violation rate V as a function of K for the four predictive-noise levels $\sigma_R \in \{0, 0.05, 0.10, 0.20\}$. The shaded regions represent the corresponding 95% Monte Carlo intervals. For the zero-violation configurations, V remains at the zero baseline across both noise levels and values of K , reflecting the structural preservation of the contextual veto. The same near-zero behavior is observed for the

remaining configurations except for the compensatory linear operator $A_L(0.90)$, which places the greatest emphasis on predictive evidence.

For $A_L(0.90)$, the effect of intervention capacity is particularly informative. Under the clean baseline, policy violations are already present at $K = 10$ and remain visible as K increases. As predictive noise increases, however, the onset of violations is progressively delayed: the corresponding curve shifts toward larger K values. At the maximum perturbation, $\sigma_R = 0.20$, the violation rate remains near zero for the smaller intervention capacities and begins to increase only beyond approximately $K = 50$. This behavior does not indicate improved policy compliance of the aggregation operator. Instead, it reflects a change in the composition of the cases reaching the Top- K cutoff under noisy predictive scores.

This compositional effect is made explicit in panel (e) of Figure 5, which decomposes the Top- K population for $A_L(0.90)$ into three mutually exclusive categories: *clipped non-vetoed* cases, *Q-vetoed* cases, and *other* cases. Here, a *clipped non-vetoed* case is a non-vetoed case ($Q_i > 0$) whose perturbed predictive score has been clipped to 1; the *Q-vetoed* category comprises all cases with $Q_i = 0$, and the *other* category contains the remaining cases. Each tile represents the complete Top- K set for a given combination of K and σ_R , so the three proportions sum to 100%.

In the clean condition (i.e., $\sigma_R = 0.0$, panel (e)), clipped non-vetoed cases are virtually absent, whereas *Q-vetoed* cases are already represented at small values of K and become increasingly prevalent as the intervention capacity grows. Once predictive noise is introduced, clipped non-vetoed cases enter the prioritized set and progressively displace *Q-vetoed* and *other* cases. The effect is strongest at $\sigma_R = 0.20$: clipped non-vetoed cases account for approximately 90.5% of the Top- K set already at $K = 10$. Their proportion subsequently decreases as K increases, although it does not disappear completely. From approximately $K = 60$ onward, *Q-vetoed* cases begin to reappear and their representation increases with K , reaching about 39.7% at $K = 100$.

The composition analysis therefore explains the apparently delayed increase in V observed for $A_L(0.90)$ under strong predictive noise. Clipping concentrates perturbed predictive scores at the admissible boundary and changes which cases occupy the highest positions of the ranking. Policy-violating cases consequently do not enter the Top- K until the intervention capacity is sufficiently large for the *Q-vetoed* cases to re-emerge among the selected alternatives. The V -versus- K analysis thus complements the noise-based ranking measures without introducing a new source of uncertainty: it reveals how intervention capacity and score perturbation jointly affect the composition of the selected set.

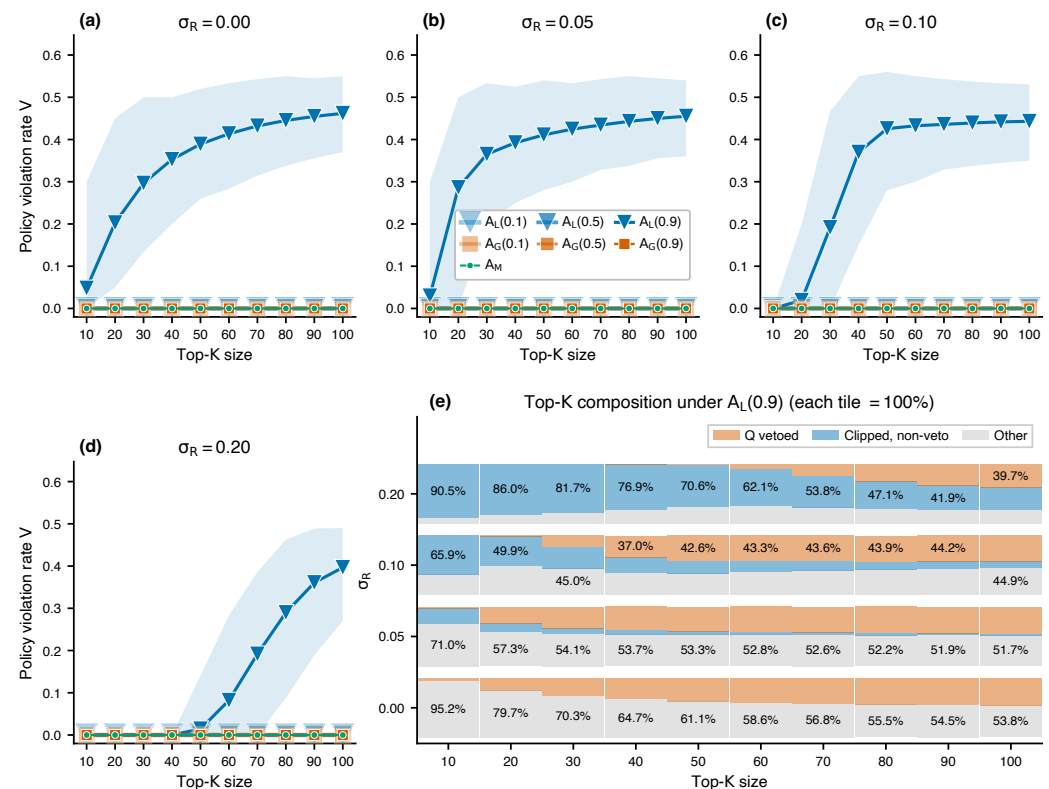


Figure 5. Sensitivity of policy compliance to intervention capacity and predictive uncertainty. Panels (a)–(d) show the mean policy violation rate V as a function of Top-K size for $\sigma_R \in \{0, 0.05, 0.10, 0.20\}$, with shaded 95% Monte Carlo intervals for the seven ranking configurations. Panel (e) shows the corresponding Top-K composition for $A_L(0.90)$ as a function of K and σ_R , with each tile representing the percentage of *clipped non-vetoed*, *Q-vetoed*, and other cases in the selected set.

Taken together, the sensitivity analyses reveal two complementary consequences of predictive uncertainty. First, noise can substantially reduce ranking stability, particularly for configurations that assign greater weight to predictive evidence. Second, its effect on policy compliance depends on both aggregation semantics and intervention capacity. Zero absorption preserves contextual vetoes independently of the predictive perturbation, whereas the compensatory linear operator can admit vetoed cases. For $A_L(0.90)$, the apparent delay in policy violations under strong noise is explained by the changing composition of the Top-K set, driven in part by clipping at the boundaries of the predictive-score domain. Thus, uncertainty changes the stability and composition of the resulting prioritization, but does not alter the underlying aggregation semantics that determine whether contextual vetoes are structurally preserved.

5.4. Case-Study Data and CADEMAs-ML Scores

The case study revisits the employee-attrition application originally introduced in CADEMAs-ML [10]. The cohort consists of $n = 100$ employees sampled from the *IBM HR Analytics Employee Attrition and Performance* dataset.¹ The present study reconstructs the original CADEMAs-ML workflow to obtain the employee-level predictive and contextual scores used in the subsequent aggregation analysis. The observed attrition label is retained only as an external reference and is not used in the computation of either score or in the prioritization process.

¹ Available at <https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset>.

The predictive score R_i is obtained from four independent machine learning ADM units representing different organizational perspectives: *Culture and Wellbeing*, *Finance*, *Operations and Development*, and *Human Resources*. Following the original application, each ADM unit is trained independently using its corresponding feature subset and H2O AutoML [24], with an 80/20 train-validation split and a one-hour time budget. The resulting models are exported as MOJO artifacts and applied to the case-study cohort to obtain positive-class attrition probabilities $\hat{p}_{ik}$. The four predictions are then combined into the cooperative predictive score

$$R_i = \sum_{k=1}^4 w_k \hat{p}_{ik}, \quad (19)$$

where the weights are proportional to the validation AUC of the respective ADM unit,

$$w_k = \frac{\text{AUC}_k}{\sum_{\ell=1}^4 \text{AUC}_\ell}. \quad (20)$$

Thus, $R_i \in [0, 1]$ represents the integrated predictive assessment of employee attrition risk while preserving the contributions of the four partially independent ADM units.

The contextual score Q_i is computed independently from the predictive layer through the *Digital/Technological Transformation* context defined in the original application. This context comprises seven fuzzy atomic rules concerning employee characteristics relevant to technological transformation. These rules are combined into four higher-level criteria: *upskilling capacity*, *career scalability*, *investment horizon*, and *role adaptability*. The resulting criteria are aggregated according to the context policy and combined conjunctively using the Gödel minimum. Consequently, failure to satisfy any of the four higher-level requirements yields $Q_i = 0$. The resulting $Q_i \in [0, 1]$ therefore represents the degree of contextual alignment of employee i with the transformation scenario, independently of the predictive attrition assessment.

The resulting cohort contains 35 employees with $Q_i = 0$, corresponding to contextual vetoes, while the remaining employees have strictly positive contextual scores. The predictive scores span almost the entire admissible range, approximately $R_i \in [0, 0.99]$, whereas the non-vetoeed contextual scores occupy a narrower range, approximately $Q_i \in [0.06, 0.73]$. This combination provides a heterogeneous decision population in which predictive risk and contextual alignment are only partially aligned. In particular, high predictive risk does not necessarily imply high contextual alignment, while contextual vetoes may occur at different levels of predictive risk. This makes the cohort suitable for examining whether the aggregation rule can alter prioritization when the underlying predictive and contextual assessments are held fixed.

5.5. Experimental Setup

The analysis considers a fixed intervention capacity of $K = 10$, corresponding to 10% of the $n = 100$ employee cohort. For each employee, the same predictive and contextual scores (R_i, Q_i) described above are combined using the seven integration configurations considered throughout the experiments: $A_L(0.1)$, $A_L(0.5)$, $A_L(0.9)$, $A_G(0.1)$, $A_G(0.5)$, $A_G(0.9)$, A_M . Here, $A_M = \min(R, Q)$ is the zero-absorbing minimum operator, whereas A_L and A_G represent the linear and weighted geometric operators, respectively. Thus, the predictive and contextual assessments, as well as all employee attributes, remain unchanged across configurations; only the integration semantics used to obtain P_i vary.

For each configuration, employees are ordered by decreasing prioritization score P_i , and the first K positions define the intervention set. Exact ties are resolved deterministically by first comparing the predictive score R_i and then using the original cohort order as a final

tie-break. This convention ensures that every configuration produces a unique Top- K set and avoids introducing an additional policy-specific rule for tied prioritization scores.

The analysis compares both the resulting score distributions and the composition of the Top-10 sets. In particular, $A_L(0.5)$ is used as a reference configuration for the rank-trajectory analysis, allowing the effect of changing the aggregation semantics to be expressed as rank displacements for employees initially selected under the reference policy. Set-level differences are additionally quantified through the intersection and Jaccard similarity between the corresponding Top-10 sets.

For brevity, implementation-level details of the case-study construction are not reproduced here. The complete workflow, including the four predictive ADM units, fuzzy-rule-based contextual evaluation, computation of R_i and Q_i , all seven aggregation configurations, deterministic ranking procedure, and generation of the reported results, is implemented in the repository accompanying this paper. The implementation therefore provides a reproducible specification of the complete case-study pipeline while allowing the present section to focus on the consequences of the aggregation choices.

5.6. Results and Interpretation

Figure 6 provides an overview of the resulting decision population and prioritization scores. Panel (a) shows that predictive risk and contextual alignment are only partially aligned in the cohort. In particular, the population contains a substantial group of contextually vetoed employees with $Q_i = 0$, while several employees with relatively high predictive risk exhibit only moderate contextual alignment. This separation between the two dimensions is important because it creates a setting in which the integration rule can materially affect the resulting priority rather than merely rescaling a common ordering.

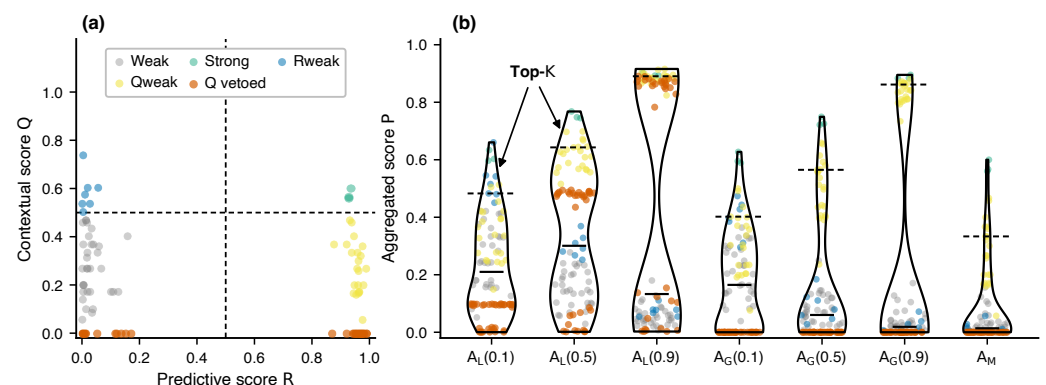


Figure 6. Employee-attrition case study ($n = 100$, $K = 10$). (a) Cohort in the (R, Q) plane, with contextually vetoed cases ($Q_i = 0$) along the horizontal axis. (b) Aggregated-score distributions for the seven integration configurations. The dashed line on each violin is the Top- K cutoff ($K = 10$); overlaid points use the same stratum colours as panel (a).

Panel (b) shows the distributions of P_i produced by the seven configurations. The Top-10 cutoff varies considerably across operators, from approximately 0.33 for A_M to approximately 0.89 for $A_L(0.9)$. This variation reflects not only differences in score scale but also changes in the relative position of employees within the ranking. In particular, increasing λ in the linear operator progressively emphasizes predictive risk and shifts the prioritization toward employees with high R_i . This effect is especially consequential for $A_L(0.9)$: one employee with $Q_i = 0$ enters the Top-10, despite being contextually vetoed. In contrast, the zero-absorbing operators A_G and A_M assign $P_i = 0$ whenever $Q_i = 0$ and therefore exclude all vetoed employees from the Top-10 in this cohort. The figure

consequently illustrates at the application level the structural distinction established in the theoretical and simulation analyses.

The distributional differences translate directly into changes in the composition of the intervention set. To make these changes explicit, Figure 7 tracks the ten employees selected under $A_L(0.5)$ as the reference configuration. Under this policy, the selected employees are characterized by high predictive scores and non-zero, generally moderate-to-high contextual scores. Changing only the aggregation rule can nevertheless move several of these employees out of the intervention capacity. The Top-10 intersection with the reference set is only 5 employees for $A_L(0.1)$ and 6 for $A_L(0.9)$, corresponding to Jaccard similarities of 0.33 and 0.43, respectively. Thus, both stronger context weighting and stronger predictive weighting can produce substantial replacement of intervention slots relative to the intermediate linear policy.

The effect is particularly pronounced for the high-predictive linear configuration. Several employees who occupy the middle or lower part of the reference Top-10 are strongly demoted under $A_L(0.9)$, while employees with very high R_i and comparatively weaker contextual alignment move upward. For example, Andrew Ross moves from rank 4 to rank 17 and Caroline Bennett from rank 5 to rank 14, whereas Noah Murphy moves from rank 7 to rank 2 and Christian Bennett from rank 9 to rank 3. The most pronounced displacement is observed for Madison Foster, whose rank changes from 10 to 37. These movements illustrate that the effect is not simply a change in the absolute scale of P_i ; the aggregation semantics alter the relative ordering of employees and consequently which employees receive intervention.

The zero-absorbing configurations are comparatively stable around the intermediate weighting. The Top-10 Jaccard similarity reaches 0.82 for both $A_G(0.5)$ and A_M , with most employees from the $A_L(0.5)$ reference set retaining positions within or close to the intervention boundary. Nevertheless, even these operators can replace individual employees when their aggregation semantics differ sufficiently from the reference linear policy. Hence, zero absorption should not be interpreted as implying identical rankings; its structural guarantee concerns the treatment of contextual vetoes, not complete rank invariance.

Taken together, the case study demonstrates that aggregation semantics can have a direct operational consequence in a realistic CADEMAs-ML workflow. The predictive and contextual assessments remain fixed throughout the comparison, yet changing only the integration operator alters the distribution of prioritization scores, the relative ranking of employees, and, ultimately, the composition of a resource-constrained intervention set. The case study therefore provides a concrete application-level illustration of the central result of the preceding analyses: the choice of aggregation policy is not a cosmetic modeling decision, because it can determine which individuals receive scarce intervention resources.

Figure 7 complements the distributional view by tracking the same Top-K intervention set under changes of aggregation semantics. The trajectories are those of the ten employees selected under $A_L(0.5)$; at the reference column the markers report absolute ranks, whereas the remaining columns report the signed rank displacement relative to that baseline (solid markers indicate an unchanged position). The shaded band marks the Top-10 region, so departures below that band correspond to employees that leave the intervention set under the corresponding operator.

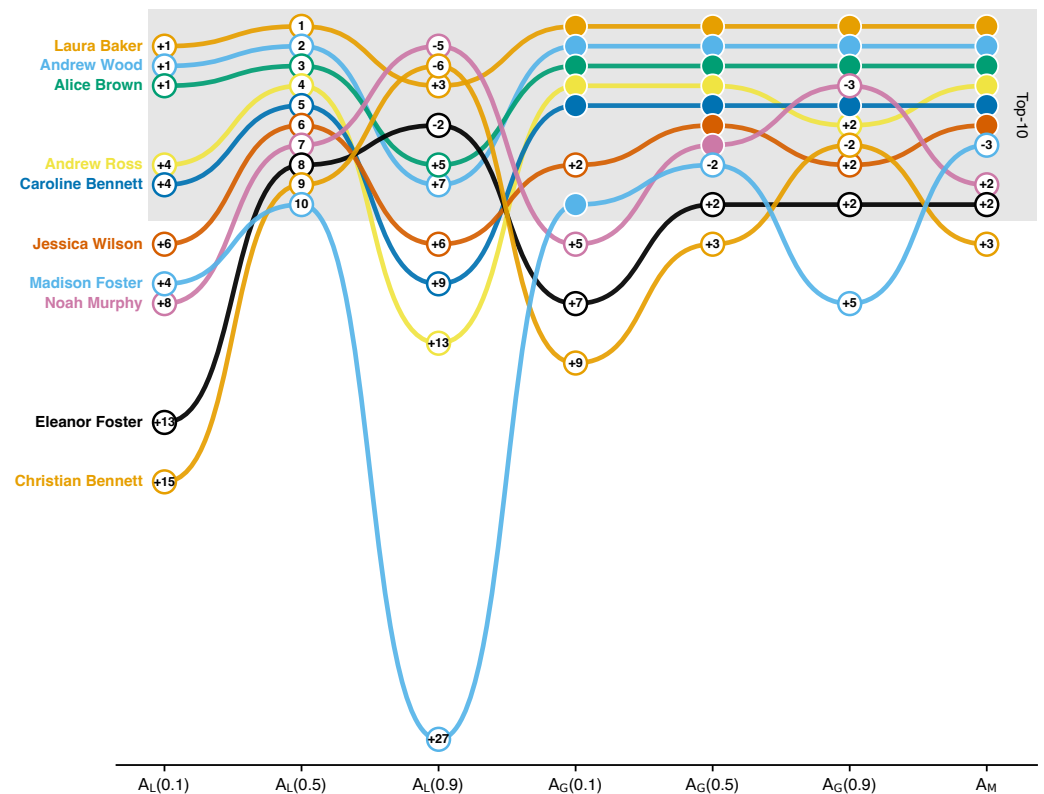


Figure 7. Rank trajectories of the Top-10 employees under $A_L(0.5)$ across the seven aggregation configurations ($n = 100$, $K = 10$). Absolute ranks are shown at the reference column $A_L(0.5)$; elsewhere, markers report the signed change relative to that baseline ($\Delta = \text{rank} - \text{rank}_{A_L(0.5)}$), with solid series-coloured markers when the rank is unchanged. The shaded band highlights the Top-10 region.

6. Discussion

The results support a central interpretation of this study: in a cooperative automated decision-making architecture, the aggregation mechanism is not a neutral post-processing choice. When predictive evidence and contextual requirements are represented by separate decision pathways, the operator that integrates them determines how these two forms of evidence can interact and, consequently, which alternatives become eligible for prioritization. This effect is visible at three complementary levels in the present study: the formal properties of the aggregation functions, the population-level behavior observed in the Monte Carlo experiments, and the composition of the intervention set in the employee-attribution case study.

The results should nevertheless not be interpreted as establishing a universal ordering of aggregation operators. The geometric and minimum operators are not shown to be generally superior to linear aggregation. Rather, the experiments demonstrate that different aggregation semantics encode different policy assumptions. In particular, zero absorption makes a contextual veto structurally non-compensable, whereas the linear operator allows predictive evidence to compensate for a null contextual assessment. The practical consequences of selecting one semantics over another depend therefore on the intended role of contextual information in the decision process.

6.1. Implications for Cooperative Automated Decision Making

The main implication for cooperative automated decision making is that aggregation should be treated as an explicit architectural and policy component rather than as an

implementation detail. In CADEMAs-ML, the predictive and contextual pathways can produce the same pair of scores (R_i, Q_i) , yet different aggregation rules can transform that identical evidence into substantially different prioritization orders. The simulations show that this difference is particularly consequential when contextual requirements represent exclusionary policies. Under zero-absorbing operators, $Q_i = 0$ is propagated to $P_i = 0$, so predictive evidence cannot restore an alternative to the prioritization set. Under the compensatory linear operator, in contrast, a sufficiently large predictive score can produce a positive aggregate score even when $Q_i = 0$. Thus, the distinction between a contextual requirement acting as a hard constraint or as a compensable preference is encoded directly in the aggregation semantics.

This observation is relevant to cooperative architectures because the different subsystems need not have the same epistemic or organizational status. A predictive ADM may estimate an outcome such as employee attrition, while a contextual subsystem may encode organizational requirements, eligibility conditions, or policy constraints. Treating both outputs simply as commensurable numerical evidence and averaging them implicitly assumes that sufficiently strong evidence from one pathway may compensate for a deficit in the other. Such an assumption may be appropriate for some decisions, but it is not policy-neutral. The choice of aggregation operator therefore determines a form of interaction between the participating decision components.

The experimental results also show that policy compliance and ranking stability are related but distinct properties. Zero absorption guarantees the preservation of contextual vetoes, but it does not imply that all non-vetoed alternatives receive similar ranks under different operators. Likewise, agreement between two operators does not imply that they implement the same policy semantics. Configurations can produce similar Top- K sets in some regions of the parameter space while differing fundamentally in their behavior at a contextual boundary. This distinction is important when evaluating cooperative decision systems: similarity of observed decisions in a particular population should not be confused with equivalence of the underlying integration policies.

The sensitivity analysis reinforces this interpretation. Predictive-score uncertainty can substantially modify rankings, especially when predictive evidence receives a high weight, but it does not remove the structural distinction between compensatory and zero-absorbing aggregation. In particular, perturbations of R do not cause zero-absorbing operators to admit alternatives with $Q_i = 0$. The observed degradation in ranking stability should therefore be understood as uncertainty acting through an already specified aggregation semantics, rather than as uncertainty eliminating the policy distinction between operators.

The employee-attrition case study provides the corresponding application level interpretation. The four predictive ADM outputs and the contextual assessment are held fixed while only the aggregation rule is changed. The resulting changes in the Top-10 composition demonstrate that the aggregation policy can determine which individuals receive scarce intervention resources. This is particularly relevant in resource constrained settings, where small changes in the ordering around the Top- K boundary can translate into actual differences in intervention allocation. The case study therefore illustrates the operational significance of the formal properties established earlier without being treated as an independent validation of the theoretical results.

More generally, these findings suggest that aggregation policies in cooperative ADM systems should be selected according to the intended relationship between evidence sources and documented policy requirements. Where contextual information represents a strict eligibility condition, zero-absorbing or otherwise non-compensatory aggregation may be appropriate. Where predictive and contextual evidence are legitimately substitutable, a compensatory operator may instead be justified. The important point is that this choice

should be explicit and auditable rather than being inherited implicitly from a convenient averaging rule.

6.2. Relation to Fuzzy MCDM and Aggregation Theory

The contribution of this work should be distinguished carefully from a contribution to aggregation theory or to fuzzy multi-criteria decision making in isolation. The operators studied here—including weighted arithmetic, weighted geometric, and minimum aggregation—are classical aggregation functions, and their basic mathematical properties are not new. Likewise, fuzzy MCDM has an extensive literature on aggregation, compensation, conjunctive decision rules, and the representation of criteria with different degrees of importance. The present work does not propose a new aggregation function, nor does it claim a new general theory of fuzzy aggregation.

The contribution instead lies in examining these familiar operators as alternative *integration semantics* within a cooperative automated decision-making architecture. The two aggregated quantities in this study are not simply homogeneous decision criteria of a conventional MCDM problem. R_i represents predictive evidence produced by an automated decision-making pathway, whereas Q_i represents contextual alignment derived from a separate knowledge- and policy-oriented pathway. Their integration therefore determines how evidence originating from different decision mechanisms interacts at the point where the final prioritization is produced.

This distinction matters because the same aggregation function can have a different governance interpretation depending on what its inputs represent. A weighted arithmetic mean used to combine two conventional performance criteria expresses a particular preference model. Applied to predictive risk and a contextual veto, the same mathematical function additionally determines whether policy restrictions can be compensated by predictive evidence. In the present setting, the algebraic property of zero absorption therefore acquires a direct decision-governance interpretation: it specifies whether a null contextual assessment is capable of being overridden at the integration stage.

From this perspective, the theoretical analysis contributes a way of connecting aggregation properties to CADEMAs-level decision behavior. Zero absorption yields contextual veto preservation, while compensability characterizes the extent to which one pathway can offset the other within the admissible score range. The simulation results then establish that these properties are not merely algebraic distinctions: they produce measurable differences in policy violation, ranking agreement, and Top-K composition. The case study further shows that the same mechanism remains visible when R_i and Q_i originate from an actual CADEMAs-ML workflow.

The relationship with fuzzy MCDM is consequently complementary rather than competitive. Fuzzy MCDM provides a mature body of aggregation models and decision rules for representing graded information and preference structures. The present study takes a subset of such aggregation semantics and asks a different question: what are the consequences when these semantics are used to integrate an automated predictive assessment with a contextual policy assessment inside a cooperative ADM architecture? This perspective also explains why the paper does not attempt to identify a universally optimal operator. The appropriate aggregation semantics depend on whether contextual information is intended to act as a preference, a soft constraint, or a strict veto.

The distinction is also important for interpreting the broader literature on advanced aggregation mechanisms such as OWA, Choquet integrals, and consensus-based aggregation. Such mechanisms provide richer ways of representing interactions, importance weights, or non-additive preferences, but adopting a more expressive operator does not by itself resolve the policy question addressed here. A more complex operator must still make

explicit whether and under which conditions contextual requirements can be compensated by predictive evidence. The present framework therefore provides a semantic criterion against which alternative aggregation mechanisms can subsequently be analyzed within CADEMAs.

6.3. Limitations

Several limitations delimit the scope of the conclusions. First, the theoretical analysis focuses on three representative aggregation regimes: linear, weighted geometric, and minimum aggregation. These operators span different degrees of compensability and provide a useful basis for isolating the role of zero absorption, but they do not exhaust the space of possible aggregation mechanisms. The conclusions should therefore not be generalized to all nonlinear, fuzzy, or non-additive operators.

Second, the Monte Carlo experiments use controlled population designs. The contextual vetoes and the distributions of (R_i, Q_i) are deliberately constructed to expose conflicts between predictive evidence and contextual requirements. This design is appropriate for testing the theoretical properties of the operators, but it does not reproduce the full complexity of deployed decision environments. In particular, the empirical threshold at which violations emerge for the linear operator is a property of the considered population and Top- K setting, not a universal critical value of λ .

Third, the uncertainty analysis considers perturbations of the predictive score R . For the operators studied here, the corresponding contextual perturbation can be obtained by exchanging the inputs and using the complementary weighting parameter for the λ -dependent operators. This symmetry avoids a redundant Monte Carlo experiment, but it does not capture all forms of uncertainty that may arise in a real contextual reasoning subsystem, such as rule uncertainty, changes in membership functions, or model-structure uncertainty.

Fourth, the employee-attribution analysis is an illustrative case study based on a cohort of 100 employees and a single contextual scenario. Although the predictive and contextual pathways are reconstructed from the original CADEMAs-ML application and the complete workflow is reproducible, the case study does not establish external validity across organizations, datasets, or intervention policies. The observed changes in the Top-10 set should therefore be interpreted as evidence that aggregation semantics can matter in a realistic workflow, rather than as evidence that one operator is generally preferable for employee-attribution interventions.

Finally, the present study evaluates the aggregation stage while treating the predictive and contextual scores as given inputs to that stage. It does not address how organizations should determine the substantive validity of the predictive models, the appropriateness of the contextual rules, or the fairness and acceptability of the resulting interventions. These are important dimensions of automated decision making, but they concern different layers of the CADEMAs architecture. Separating them from the aggregation analysis is necessary to identify the specific contribution of this work, while also limiting the conclusions that can be drawn from the reported experiments.

7. Conclusions and Future Work

This study investigated how aggregation semantics affect prioritization in Cooperative Automated Decision-Making Systems when predictive evidence and contextual requirements are integrated into a single decision score. The analysis considered three representative aggregation regimes—linear, weighted geometric, and minimum aggregation—and examined their structural properties, policy-compliance behavior, sensitivity to predictive uncertainty, and consequences for resource-constrained Top- K prioritization.

The results provide a direct answer to the research questions. Regarding **RQ1**, the analysis shows that zero absorption is the key structural property determining whether a contextual veto is preserved by the integration stage. The weighted geometric and minimum operators satisfy this property, whereas the linear operator with $\lambda \in (0,1)$ is compensatory and can assign a positive prioritization score to an alternative with $Q_i = 0$. Thus, contextual veto preservation is not an automatic consequence of combining predictive and contextual scores; it depends on the aggregation semantics.

For **RQ2**, the controlled experiments show that this structural difference has direct consequences for policy compliance. Under the considered population designs and Top-K selection conditions, the zero-absorbing operators preserve the contextual veto mechanism, whereas the linear operator can introduce vetoed alternatives into the prioritized set. The magnitude of this effect depends on the predictive weighting and on the distribution of predictive and contextual evidence. These results should therefore be interpreted as a characterization of the considered conditions rather than as evidence that one aggregation family is generally superior to another.

For **RQ3**, the sensitivity analysis demonstrates that predictive uncertainty can substantially modify prioritization, particularly when predictive evidence receives greater weight. Increasing perturbations of R reduce agreement with the corresponding clean rankings and can alter the composition of the Top-K set. The effect is therefore not limited to small numerical changes in aggregate scores: uncertainty can propagate to the actual alternatives selected for intervention. At the same time, the structural policy distinction between compensatory and zero-absorbing operators remains intact under the considered perturbations.

For **RQ4**, the results indicate that uncertainty and aggregation semantics interact rather than becoming indistinguishable under noise. Increasing predictive perturbations can reduce agreement among different configurations, but the resulting rankings continue to reflect the underlying integration rules. The additional analysis over different intervention capacities shows that the observed policy-violation behavior also depends on the Top-K boundary. In particular, under strong predictive noise, clipping of perturbed predictive scores can substantially alter the composition of the selected set and thereby modify the observed violation rate. This highlights that policy compliance should be evaluated jointly with the uncertainty model and the operational intervention capacity, rather than solely from the aggregate function in isolation.

The employee-attrition case study provides an application-level illustration of these findings. Using the same predictive and contextual assessments for all configurations, changing only the aggregation semantics produces different prioritization distributions and can replace several employees within a fixed Top-10 intervention budget. The case study therefore demonstrates the practical relevance of the theoretical distinction: aggregation is capable of changing not only the numerical priority assigned to an employee but also whether that employee receives a scarce intervention opportunity.

Taken together, the findings support a more explicit treatment of aggregation as a policy-bearing component of cooperative automated decision-making architectures. The contribution is not a new aggregation function, nor a claim that nonlinear or non-compensatory operators are universally preferable. Rather, the study provides a formal and empirical characterization of how established aggregation semantics translate the relationship between predictive evidence and contextual requirements into prioritization behavior within CADEMAs-ML. This perspective makes the choice of integration rule an explicit design and governance decision that can be analyzed in terms of properties such as zero absorption, compensability, policy compliance, and ranking stability.

Several directions follow naturally from this work. First, future studies can extend the analysis to richer aggregation families, including ordered weighted averaging (OWA), Choquet integrals, and other non-additive operators, to determine how interaction and importance structures affect contextual veto preservation and compensation. Second, aggregation parameters could themselves be learned from historical decisions or stakeholder preferences, leading to learned aggregation mechanisms whose policy properties can be analyzed before deployment. Third, future CADEMAs implementations could consider dynamic aggregation policies in which the integration rule or its parameters adapt to changes in context, uncertainty, resource availability, or decision objectives. Finally, extending the evaluation to larger and heterogeneous real-world cohorts would allow the interaction between aggregation semantics, uncertainty, and intervention constraints to be studied beyond the controlled and illustrative settings considered here.

Author Contributions: To be completed.

Funding: This research was supported by the project “Study, Analysis and Evaluation of Cooperative Automated Decision-Making Systems (CADEMAS)”, PID2023-146575NB-I00, funded by MCIU/AEI/I0.13039/501100011033 and by FSE+.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: To be completed.

Acknowledgments: Mariia Godz has a pre-doctoral research contract associated with the project “Study, Analysis and Evaluation of Cooperative Automated Decision-Making Systems (CADEMAS)”, PID2023-146575NB-I00, funded by MCIU/AEI/I0.13039/501100011033 and by FSE+.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Teng, T.H.; Tan, A.H. Cognitive Agents Integrating Rules and Reinforcement Learning for Context-Aware Decision Support. In Proceedings of the Proceedings of the 2008 IEEE/WIC/ACM International Conference on Web Intelligence and Intelligent Agent Technology, 2008, pp. 318–321. <https://doi.org/10.1109/WIIAT.2008.163>.
2. Yao, W.; Kumar, A. Integrating Clinical Pathways into CDSS Using Context and Rules: A Case Study in Heart Disease. In Proceedings of the Proceedings of the 2nd ACM SIGHIT International Health Informatics Symposium, 2012, pp. 267–276. <https://doi.org/10.1145/2110363.2110431>.
3. Abu-Rasheed, H.; Abdulsalam, M.; Weber, C.; Fathi, M. Supporting Student Decisions on Learning Recommendations: An LLM-Based Chatbot with Knowledge Graph Contextualization for Conversational Explainability and Mentoring. *Frontiers in Artificial Intelligence* **2024**. <https://doi.org/10.35542/osf.io/ervym>.
4. Casado, D.F.; Fernandez, R.P. A Governance-Aware Methodological Framework for Cognitive AI Integration in Marine CAD: A Conceptual Pipe Routing Use Case. *Journal of Marine Science and Engineering* **2025**, *14*, 1352. <https://doi.org/10.3390/jmse14011352>.
5. Cronholm, S.; Göbel, H. Design Principles for Human-Centred AI. In Proceedings of the ECIS 2022 Research Papers, 2022. Research paper 32.
6. Fanfani, M.; Palesi, L.A.I.; Nesi, P. Human-Centered AI for Decision Support Systems: A Systematic Review of Application Domains, Architecture Designs, Current Trends and Future Directions. *Big Data and Cognitive Computing* **2025**, *10*, 186. <https://doi.org/10.3390/bdcc10060186>.
7. Smirnov, A.V.; Levashova, T.; Ponomarev, A.; Shilov, N. Methodology for Multi-Aspect Ontology Development: Ontology for Decision Support Based on Human-Machine Collective Intelligence. *IEEE Access* **2021**, *9*, 130721–130739. <https://doi.org/10.1109/ACCESS.2021.3116870>.
8. Marot, A.; Kelly, A.; Naglic, M.; Barbesant, V.; Cremer, J.L. Perspectives on Future Power System Control Centers for Energy Transition. *Journal of Modern Power Systems and Clean Energy* **2022**, *10*, 328–344. <https://doi.org/10.35833/mpce.2021.000673>.
9. Novoa-Hernández, P.; Pelta, D.; Godz, M.; Verdegay, J.; Buendia-Carrillo, D. CADEMAs – A Framework for Cooperative Automated Decision-Making Systems. In Proceedings of the Proceedings of the 2026 IEEE Conference on Artificial Intelligence (CAI), 2026, pp. 756–761. <https://doi.org/10.1109/CAI68641.2026.11536392>.

10. Godz, M.; Novoa-Hernández, P.; Pelta, D. A Cooperative and Context-Aware Approach for Employee Attrition Prevention. In *Information Processing and Management of Uncertainty in Knowledge-Based Systems*; Springer Nature Switzerland, 2026; pp. 203–217. https://doi.org/10.1007/978-3-032-29000-7_15.
11. Yager, R. On Ordered Weighted Averaging Aggregation Operators in Multicriteria Decisionmaking. *IEEE Transactions on Systems, Man, and Cybernetics* **1988**, *18*, 183–190. <https://doi.org/10.1109/21.87068>.
12. Yager, R.R. Extending Multicriteria Decision Making by Mixing t-Norms and OWA Operators. *International Journal of Intelligent Systems* **2005**, *20*, 453–474. <https://doi.org/10.1002/int.20075>.
13. Mardani, A.; Nilashi, M.; Zavadskas, E.K.; Awang, S.; Zare, H.; Jamal, N.M. Decision making methods based on fuzzy aggregation operators: Three decades review from 1986 to 2017. *International Journal of Information Technology & Decision Making* **2018**, *17*, 391–466. <https://doi.org/10.1142/S021962201830001X>.
14. Csiszár, O. Ordered Weighted Averaging Operators: A Short Review. *IEEE Systems, Man, and Cybernetics Magazine* **2021**, *7*, 4–12. <https://doi.org/10.1109/MSMC.2020.3036378>.
15. del Moral, M.J.; Chiclana, F.; Tapia García, J.M.; Herrera-Viedma, E. An Analysis on Consensus Measures in Group Decision Making. In Proceedings of the Proceedings of the 2017 4th International Conference on Control, Decision and Information Technologies (CoDIT), 2017, pp. 0262–0267. <https://doi.org/10.1109/CoDIT.2017.8102605>.
16. Trillo, J.R.; Cabrerizo, F.J.; del Moral, M.J.; Morente-Molinera, J.A.; Tapia, J.M.; Herrera-Viedma, E. A Consensus-Based Multi-Criteria Group Decision-Making Method Based on an Aggregated Operator Customised by Experts. In Proceedings of the Proceedings of the 2023 9th International Conference on Control, Decision and Information Technologies (CoDIT), 2023. <https://doi.org/10.1109/CoDIT58514.2023.10284209>.
17. Yang, Y.W.; Lin, J.K.; Fu, Y.; Luo, Y. Tolerance Framework for Robust Group Multiple Criteria Decision Making. *Expert Systems with Applications* **2022**, *204*, 117430. <https://doi.org/10.1016/j.eswa.2022.118208>.
18. Liao, Z.; Liao, H.; Zhang, X. A Contextual Choquet Integral-Based Preference Learning Model Considering Both Criteria Interactions and the Compromise Effects of Decision-Makers. *Expert Systems with Applications* **2023**, *213*, 118977. <https://doi.org/10.1016/j.eswa.2022.118977>.
19. Stroppiana, D.; Boschetti, M.; Brivio, P.A.; Manfron, G.; Antoninetti, M. Explainable Multi-Criteria Data-Driven Environmental Status Assessment from Remote Sensing. In Proceedings of the Proceedings of the IEEE MELECON 2022, 2022. <https://doi.org/10.1109/MELECON53508.2022.9842919>.
20. Alcantud, J.C.R. Multi-Attribute Group Decision-Making Based on Intuitionistic Fuzzy Aggregation Operators Defined by Weighted Geometric Means. *Granular Computing* **2023**, *8*, 1857–1866. <https://doi.org/10.1007/s41066-023-00406-w>.
21. Merkert, J.; Mueller, M.; Hubl, M. A Survey of the Application of Machine Learning in Decision Support Systems. In Proceedings of the Proceedings of the 23rd European Conference on Information Systems (ECIS), 2015. <https://doi.org/10.18151/7217429>.
22. Pérez-Cañedo, B.; Porras, C.; Pelta, D.; Verdegay, J. Modeling Contexts as Fuzzy Propositions in Optimization Problems. *IEEE Transactions on Fuzzy Systems* **2023**, *31*, 1474–1483. <https://doi.org/10.1109/tfuzz.2022.3203786>.
23. Pérez-Cañedo, B.; Novoa-Hernández, P.; Porras, C.; Pelta, D.; Verdegay, J. Contextual analysis of solutions in a tourist trip design problem: A fuzzy logic-based approach. *Applied Soft Computing* **2024**, *154*, 111351. <https://doi.org/10.1016/j.asoc.2024.111351>.
24. H2O.ai. H2O.ai: The Open Source AI Platform. 2026. Available online: <https://www.h2o.ai/> (accessed on 12 January 2026).

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.